PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons) with research
Multidisciplinary Courses

FIRST YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2024-25

(SEMESTER: FIRST AND SECOND)

i.e

First Semester	***:***	***November/December, 2024***
Second Semester	***:***	***April/May, 2025***

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Agriculture

Semester-I

Course Title: AGRICULTURAL BOTANY AND CROP PHYSIOLOGY
Course Code: MDC-1

Objective:
To learn ecological concepts and theories of evolution, growth, and development of crops.

Max. Marks:75			**Credit hours:2+1=3**
1.	**Theory:50**	**Written Exam:40**	**Internal Assessment:10**
2.	**Practical:25**	**Written Exam:20**	**Internal Assessment:5**

Period per Week:
1. **Theory:** Three of 45 minutes duration (Two Hours)
2. **Practical:** One of two hours duration (Two Hours)

INSTRUCTIONS FOR PAPER SETTER:
Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and will comprise 8 short-answer type questions of one mark each, covering the whole syllabus. Units 'I', 'II', 'III', and 'IV' will have two questions each, and one question is to be attempted out of each Unit.

Teaching Aids:
Teachers can use lectures, lecture cum demonstration, projects, assignments, problem-solving, inductive and deductive methods, and may use L.C.D., L.E.D., multimedia projectors in addition to black/whiteboards.

Unit-I

Plant ecology, nature and scope of ecology, concept of environment. Ecological factors and their influence on the distribution of vegetation. Main ecological plant communities and their adaptations. Concept of ecosystem and biosphere.

Unit-II

Sources of pollution, environmental conservation and health. Concept of the center of origin of crop plants. Domestication of crops and evaluation of germplasm. Weed ecology and weed competition.

Unit-III

Introductory concept of theories of organic evolution. Evolution of important groups of plants. Variation, their types and causes, adaptations. Growth and development of crops: Photosynthesis, respiration, net photosynthesis, translocation of assimilates.

Unit-IV

Growth of plants: developmental stages, physiological growth components, factors influencing growth and development. Plant growth substances, growth expressions. Transpiration, absorption of food material by plants. Water absorption and translocation.

Practical:
Growth analysis experiments demonstrating various aspects of respiration, photosynthesis, and ascent of sap. Water culture techniques/deficiency and toxicity of major and minor elements.

Suggested Readings:

1. Devlin, R.M., *Plant Physiology*, Prindle Weber & Schmidt Publisher, New York, 1983.
2. Kochhar, P.L., *Plant Physiology*, Trumen Publishers, Jalandhar, 2010.
3. Bhatia, K.N., and Widge, R., *Foundation of Botany*, Truman Publishers, Jalandhar, 2010.

Semester-II

Course Title: AGRICULTURAL MICROBIOLOGY
Course Code: MDC-2

Objectives:
To learn various applications of microorganisms in agriculture, industry, and environment.

Max. Marks:75			**Course Credits = 2+1=3**
1.	**Theory:50**	**Written Exam: 40**	**Internal assessment:10**
2.	**Practical:25**	**Written Exam: 20**	**Internal assessment:5**

Periods per Week:
1. Theory: Three of 45 minutes duration (Two Hours)
2. Practical: One of two hours duration (Two Hours)

INSTRUCTIONS FOR PAPER SETTER:
Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and will comprise 8 short-answer type questions of one mark each, covering the whole syllabus. Units 'I', 'II', 'III', and 'IV' will have two questions each, and one question is to be attempted out of each Unit.

Teaching Aids:
Teachers can use lectures, lecture cum demonstration, projects, assignments, problem-solving, inductive and deductive methods, and may use L.C.D., L.E.D., multimedia projectors in addition to black/whiteboards.

Unit-I

Microbiology: Introduction, various branches of microbiology, history of soil microbiology. Classification of soil organisms: animalia, plantae, fungi, Monera.
Soil bacteria classification, soil viruses, optimum conditions for microbial activity. Functions of soil microorganisms, role of microbes in soil fertility.

Unit-II

Different approaches to study soil microbiology.
Food preservation methods: physical and chemical preservation methods.
Biological nitrogen fixation, legume-rhizobium symbiosis.
Vermicomposting: types of vermicomposting, method of vermicomposting, nutrient content of vermicompost.

Unit-III

Environmental Microbiology: Microbiology of organic compost.
Bio-filtration, purification of water, sedimentation, filtration, and disinfection.
Microbiological analysis of water purity.
Types of soil microbes, organic matter decomposition, composition of litter, carbon assimilation, and immobilization, microorganisms associated with organic matter decomposition.

Unit-IV

Organic matter dynamics in soil, factors affecting organic matter decomposition.
Biogeochemical cycle of carbon, nitrogen, phosphorus, and sulfur.
Crop protection using microorganisms: microbial herbicides, bacterial insecticides, virus insecticides, entomopathogenic fungi, microbiology of lignocellulose degradation in the rumen.

Practical:

- Methods in Microbiology – cultivation and isolation of microorganisms.
- Demonstration of techniques for pure culture of microorganisms by streak plate, pour plate, and spread plate techniques.
- Methods of culture preservation and maintenance.
- Measurement of microbial activity in soil by respiration method.
- Identification of symbiotic bacteroids of rhizobia.
- Observation of root colonization by V.A.M. fungi.
- Enzymatic test of milk by methylene blue reductase test.
- Demonstration of microorganisms in soil, water, air, and food.

Suggested Readings:

1. Dubey, R.C., and Maheshwari, D.K., *A Textbook of Microbiology*, S. Chand & Company Ltd., New Delhi, 2010.
2. Peleczar, M.J., Chain, E., and Kreis, N.R., *Microbiology*, Tata-McGraw-Hill Company, 2008.
3. Alcamo, I.E., *Fundamentals of Microbiology*, Benjamin Cummings Publishing Company, 2007.
4. Rangaswami, G., and Bagyaraj, D.J., *Agricultural Microbiology*, Prentice Hall of India Ltd., New Delhi, 2005.

Biochemistry

Semester-I

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

• Set nine questions in all. All questions carry equal marks.
• Five questions to be attempted.
• Question number one will be compulsory having 7-10 short answer types covering the whole syllabus
• Set two questions from each Section, and each question should be further divided in two to three parts. Any one question to be attempted from each section.

MDC-1-BCH: Introduction to Biochemistry (2+1) **Marks: 45 (40+10(Int.)**

Objective: To learn about biomolecules, their structure and function in the biological system

SECTION – I

(Lectures: 8)

Water: Physico chemical properties of water, dissociation and association constants. pH and buffers, pI, pka, Henderson Hasselbatch equation and its implications. **Carbohydrates**: Structure of important mono, di-, oligo- and polysaccharides, glycoproteins, peptidoglycan, glycolipids and lipopolysaccharides. Storage and cell wall polysaccharides.Enantiomers and anomers.Acid mucopolysaccharides.

SECTION- II

(Lectures: 8)

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids and terpenes, sterols, steroids: Prostaglandins.

SECTION– III

(Lectures: 8)

Proteins: Structure of amino acids, nonprotein and rare amino acids and their chemical reactions. Structural organization of proteins (primary, secondary, tertiary and quaternary domain structure), protein classification and function.Forces stabilizing primary, secondary and tertiary structure.

SECTION-IV

(Lectures: 8)

Nucleic Acids: Structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides.Structures of RNA & DNA.Porphyrins: General introduction

Suggested Books:

1. Lehninger A.L., Nelson D.L. and Cox M.M. (2005).Principles of biochemistry (W. H. Freeman, USA).
2. Stryer L, J. M. Berg, J.L. Tymoczko (2001). Biochemistry (W.H. Freeman and Company, New York).
3. Rawn J.D. (1989). Biochemistry (Neil Patterson Publishers).
4. G.L. Zubay, W.W. Parson, D.E. Vance. (1995). Principles of biochemistry: Student study art notebook (Wm.C. Brown).
5. Bucke C. (1999). Carbohydrate biotechnology protocols (Humana Press).
6. H.R. Horton, A.J. Scism, L.A. Moran, R.S. Ochs, J.D Rawn, K.G. Scrimgeour. (2006). Principles of Biochemistry (Prentice Hall).

PRACTICALS:

MDC-1-BCH: Introduction to Biochemistry **Marks: 25 (20 +5 (Int.)**

1. Preparation of biological buffers and solutions of different pH and molarity.
2. Qualitative tests for carbohydrates.
3. Qualitative tests for amino acids.
4. Estimation of saponification value of fat/oil.
5. Estimation of iodine value of fat/oil.
6. Titration curve of an amino acid and determination of its pKa and isoelectric pH.
7. Isolation of casein from milk.
8. Isolation of starch from potato/gluten from wheat.
9. Estimation of vitamin C

Semester-II

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

• Set nine questions in all. All questions carry equal marks.
• Five questions to be attempted.
• Question number one will be compulsory having 7-10 short answer types covering the whole syllabus
• Set two questions from each Section, and each question should be further divided in two to three parts. Any one question to be attempted from each section.

MDC-2-BCH: Introduction to Biochemistry (2+1) **Marks: 50(40 +10 (Int.)**

Objective: To learn about biomolecules, their structure and functional in the biological system

SECTION – I

(Lectures: 8)

Water: Physico chemical properties of water, dissociation and association constants. pH and buffers, pI, pka, Henderson Hasselbatch equation and its implications. **Carbohydrates**: Structure of important mono, di-, oligo- and polysaccharides, glycoproteins, peptidoglycan, glycolipids and lipopolysaccharides. Storage polysaccharides and cell walls.Enantiomers and anomers.Acid mucopolysaccharides.

SECTION- II

(Lectures: 8)

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids and terpenes, sterols,steroids: Prostaglandins.

SECTION – III

(Lectures: 8)

Proteins: Structure of amino acids, nonprotein and rare amino acids and their chemical reactions. Structural organization of proteins (primary, secondary, tertiary and quaternary domain structure), protein classification and function.Forces stabilizing primary, secondary and tertiary structure.

SECTION - IV

(Lectures: 8)

Nucleic Acids: Structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides.Structures of RNA & DNA.Porphyrins: General introduction

Suggested Books:

1. Lehninger A.L., Nelson D.L. and Cox M.M. (2005).Principles of biochemistry (W. H. Freeman, USA).
2. Stryer L, J. M. Berg, J.L. Tymoczko (2001). Biochemistry (W.H. Freeman and Company, New York).
3. Rawn J.D. (1989). Biochemistry (Neil Patterson Publishers).
4. G.L. Zubay, W.W. Parson, D.E. Vance. (1995). Principles of biochemistry: Student study art notebook (Wm.C. Brown).
5. Bucke C. (1999). Carbohydrate biotechnology protocols (Humana Press).
6. H.R. Horton, A.J. Scism, L.A. Moran, R.S. Ochs, J.D Rawn, K.G. Scrimgeour. (2006). Principles of Biochemistry (Prentice Hall).

PRACTICALS:

MDC-2-BCH: Introduction to Biochemistry **Marks: 25 (20 +5(Int.)**

1. Preparation of biological buffers and solutions of different pH and molarity.
2. Qualitative tests for carbohydrates.
3. Qualitative tests for amino acids.
4. Estimation of saponification value of fat/oil.
5. Estimation of iodine value of fat/oil.
6. Titration curve of an amino acid and determination of its pKa and isoelectric pH.
7. Isolation of casein from milk.
8. Isolation of starch from potato/gluten from wheat.
9. Estimation of vitamin C

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Bioinformatics

Semester-I

BINF-1004: INTRODUCTION TO BIOINFORMATICS - I

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

(Credit: 3)
Theory: 40
Int. ass: 10
Time: 3 Hours

Objective:
Introduction to Bioinformatics exposes the students to study the basics and applications of Bioinformatics.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Introduction to Bioinformatics: Definitions, History, Goal, Scope and its Applications.

Literature Database: PubMed, PMC, NCBI Book Shelf.

Biological Databases Classification: Primary, Secondary and Specialized databases.
Nucleotide Sequence Database: Genbank, EMBL, DDBJ, INSDC, IBDC and its data portals.
Protein Sequence Database: Uniprot.
Database flat file formats: Genbank/Genpept, EMBL and Uniprot flat files.
Sequence file formats: FASTA and PIR Format.
Information retrieval for biological databases: Gquery, Boolean search terms and statements

UNIT – II

Sequence Alignment:
Evolutionary basis of Alignment, Concept of Sequence Homology, Similarity, and Identity.

Pairwise sequence alignment: Concept of Global and Local alignment, Dot matrix method, Dynamic programming (Needleman Wunsch algorithm and Smith-Waterman algorithm) Concepts behind General gap and affine gap penalty.

Substitution matrices: Amino acid substitution matrices (PAM and BLOSUM)

Tools for Sequence Alignment: BLAST: Algorithm, types, understanding the BLAST output and the significance E-value and Bit score, different uses of BLAST.

FASTA algorithm, Difference between BLAST & FASTA.
Applications of Sequence Alignments.

Recommended Books:

- Stallings (2006), Operating Systems, Internals and Design Principles, 5th edition, Pearson Education, India.
- Andrew S. Tanenbaum (2007), Modern Operating Systems, 2nd edition, Prentice Hall of India, India.
- Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By
- Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.
- Bioinformatics: Sequences, structure and databanks by Des Higgins and Willie Taylor, Oxford University Press, (2000).

BINF-1054: INTRODUCTION TO BIOINFORMATICS – I (PRACTICAL)

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

Max Marks: 25 (20+5)

1. Searching literatures over PubMed.
2. Understanding and using Sequence information resources on web: EMBL-ENA, Genbank and DDBJ.
3. The information retrieval using Gquery.
4. Understanding and using Protein information resources on web: Uniprot
5. Performing BLASTp/n and interpreting its results.

Semester-II

BINF-2004: INTRODUCTION TO BIOINFORMATICS - II

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

(Credit: 3)
Theory: 40
Int. ass: 10
Time: 3 Hours

Objective:

The aim of this subject is to exposes the students to Bioinformatics sequence analysis and its applications.

General Instruction:

- The question paper will have seven questions, and each question have 16 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Multiple Sequence Alignment: SP (Sum of Pairs) measure to determine scoring function, Multiple Sequence Alignment methods: Progressive (Clustal Omega and PILEUP), Iterative and Block-based methods of MSA. Application of MSA.

Tools for MSA: Clustal Omega, Clustal-X and T-coffee.

UNIT - II

Phylogenetic analysis: Basic terminology in Phylogenetics, Introduction to phylogenetic analysis and its application, Phylogenetic tree terminology and topologies.

Methods of Phylogenetic Tree Construction:

Distance Based Methods: NJ, FM & UPGMA
Character Based Methods: Maximum Parsimony & Maximum Likelihood.
Phylogenetic Tree Evaluation Methods: Bootstrapping & Jackknifing.

Tools for Phylogenetic Analysis: Introduction to Phylip and MEGA packages.

Protein Structural databases: PDB, SCOP and CATH

Recommended Books:

- Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).

- Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By
- Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.
- Bioinformatics: Sequences, structure and databanks by Des Higgins and Willie Taylor, Oxford University Press, (2000).

BINF-2054: INTRODUCTION TO BIOINFORMATICS – II (PRACTICAL)

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

Max Marks: 25 (20+5)

1. Performing multiple sequence alignment using Clustal Omega.
2. Building Phylogenetic tree using:
    a. Distance-based & character-based methods by PHYLIP or MEGA
    b. Tree evaluation
3. Understanding the PDB and its file format.
   Downloading and visualizing protein structure from PDB

Biotechnology

Semester-I
BIOT – IDC1-T: (Foundations of Biotechnology)

TOTAL CREDITS: 3

Credits: 2

Objectives:
This course will introduce the basic concepts of biotechnology to the students.

They will learn about the history of biotechnology, the foundations of modern biotechnology, the role of biotechnology in the fermentation industry, environment, and modern medicine, and the ethical implications of biotechnology.

UNIT I

Basics of biotechnology: Green, White, Red, and Blue biotechnology.
Model organisms in research: ***E. coli, Saccharomyces cerevisiae, Drosophila melanogaster, Caenorhabditis elegans, Danio rerio, Mus musculus***.

UNIT II

Role of viruses and bacteriophages in biotechnology.
Fundamentals of recombinant DNA technology: Restriction Enzymes.
Role of biotechnology in diagnostics, introduction to gene therapy.
Transgenic animals and their applications in biotechnology.

Reference Books:

1. ***Comprehensive Biotechnology***, Murray Moo-Young, 2nd edition (2011), Pergamon Press.
2. ***Introduction to Biotechnology***, William J. Thieman & Michael A. Palladino, 3rd edition (2012), Benjamin Cummings.
3. ***Biotechnology Expanding Horizons***, B. D. Singh, 4th edition (2012), Kalyani Publishers.
4. ***The Ethics of Biotechnology (Biotechnology in the 21st Century)***, Jonathan Morris, Chelsea House Pub(L), 1st edition (2005).
5. ***Biotechnology: Applying the Genetic Revolution***, David P. Clark & Nanette J. Pazdernik, 1st edition (2008), Academic Cell.
6. ***Molecular Biotechnology***, Sandy B. Primrose, 2nd edition (1991), Blackwell Scientific Publications, ISBN.17
7. ***Biotechnology: Demystifying the Concepts***, David Bourgaize, Thomas R. Jewell & Rodolfo G. Buiser, 1st edition (1999), Benjamin Cummings.

BIOT – IDC1-T: (Foundations of Biotechnology) Practicals

Credits: 1

1. Introduction and use of Auto-pipettes, pH meter.
2. Use of centrifuges, light microscope, electrophoretic apparatus.
3. Introduction to vortex mixer, magnetic stirrer, rocker, laminar hoods, autoclave.
4. Cell disruption and cell lysis of bacterial cell (Lysozyme digestion).
5. Staining of bacteria, yeast, and fungi.
6. Isolation of microbes from the environment.
7. Handling and disposal of hazardous reagents (acids, carcinogenic chemicals like acrylamide, ethidium bromide).

Semester-II

BIOT–IDC2-T: (General Microbiology)

TOTAL CREDITS: 3

Credits: 2

Objectives:

Microbes play a significant role in understanding medical science and industries. This course aims to study the microbes from basic concepts, including understanding cell structure and the application of microbes.

UNIT I

History and scope of microbiology, principles of microscopy; resolution, magnification; compound microscope.

Morphology and structure of bacteria; growth curve, cell wall (Gram-positive and Gram-negative bacteria), flagella, cilia, chemotaxis.

Morphology of fungi and viruses.

UNIT II

Sterilization and pasteurization; Physical and chemical agents to kill microbes.

Normal microflora in humans. Types of microbial pathogens and diseases caused by them.

Microbial interactions like symbiosis and antibiosis.

Reference Books:

1. ***Cappuccino, J.G., Sherman, N.*** (2007). ***Microbiology: A laboratory manual***, (Pearson Benjamin Cummings).
2. ***Tortora, G.J., Funke, B.R., and Case, C.L.*** (2009) ***Microbiology: An introduction*** (Benjamin/Cummings publishing company, Inc).
3. ***R.Y. Stanier, M. Doudoroff, E.A. Adelberg*** (1999). ***General Microbiology*** (MacMillan Press, London).
4. ***M.J. Pelczar, E.C. Sun Chan, N.R. Krieg*** (1986). ***Microbiology*** (Tata McGraw Hill Publication, New Delhi).
5. ***H.G. Schlegel, C. Zaborosch, M. Kogut*** (1993). ***General Microbiology*** (Cambridge University Press).
6. ***S.C. Prescott, C.G. Dunn*** (1959). ***Industrial Microbiology*** (McGraw-Hill).
7. ***Purohit, S.S.*** (2003). ***Microbiology: Fundamentals and Applications*** (Agrobios, India).
8. ***Postgate, J.R.*** (2000). ***Microbes and Man*** (Cambridge University Press).

BIOT–IDC2-P: (General Microbiology) Practicals

Credit: 1

1. Sterilization of liquid and solid items.
2. Preparation of complex and synthetic media.
3. Concept of laminar flow hoods and Biosafety cabinets: principle and working.
4. Dilution and pour plating techniques.
5. Gram staining.
6. Isolation of microorganisms from soil.
7. Isolation of microbes from water.

Botany

Semester-I

Course Title: **PLANTS- IMPORTANCE AND PROPAGATION** **Course Code: MDC-1**

Objective: The course has been framed to familiarize students with the importance of plants for sustenance of our planet in general and mankind in particular. It also aims to educate them about different modes of plant reproduction.

Max. Marks : 75 **Course Credits =** 3
Theory : 75 Written Exam: 60 Internal Assessment: 15

Periods per week: Theory- Three of 1 hour duration (Three Hours)

Instructions for Paper Setters:

Question paper will have four sections. Examiner will set a total of nine questions comprising of one compulsory question consisting of MCQ and fill in the blanks covering the whole syllabus and two questions from each unit. Student will attempt **five** questions in all – one question from each unit along with one compulsory question.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., multimedia projector in addition to black/white board.

Unit – I

Diversity in habit (herb, shrub and tree) and habitat (terrestrial and aquatic), plant adaptations (hydrophytes and xerophytes).

Unit – II

Plants as source of: food (Wheat, Rice); timber (Shisham, Teak); medicine (Quinine, Sarpgandha); essential oils (Lemon Grass,Tulsi) and beverages (Tea and Coffee) - General morphology, botanical names and parts used.

Unit – III

General account of asexual means of reproduction; Natural and artificial means of vegetative propagation; advantages and limitations; propagation by bulbs, corms, tubers, rhizomes, runners, stolons and suckers – general account; propagation by cutting, layering, grafting and budding – general account.

Unit – IV

General account of sexual reproduction - structure of flower and its various forms including androecium and gynoecium; types of pollination (self vs. cross), fertilization; basic concept of dicot seed, structure and germination.

Suggested Readings:

1. Bhojwani, S.S., Bhatnagar, S.P. and Dantu, P.K. (2018). The embryology of angiosperms. (6th edition). Vikas Publishing House, Noida.
2. Coulter, J.M. (1903). Morphology of Angiosperms (Morphology of Spermatophytes. Part-II). Nebu press, pp 376.
3. Johri, B.M. and Srivastava, P.S. (2013). Reproductive Biology of Plants. Springer Science and Business Media, pp 320.
4. Khan, A.S. (2017). Flowering Plants: Structure and Industrial Products. Wiley; 1st edition, United Kingdom, pp 344.
5. Kochhar, S.L. (2016). Economic Botany. Cambridge University Press, pp 680.
6. Nanda, K.K. and Kochar V.K. (1985). Vegetative Propagation of Plants. Kalyani Publishers, New Delhi.
7. Ramawat, K.G. *et al*. (2014). Reproductive Biology of Plants. CRC Press, Boca Raton.
8. Sadhu, M.K. (1999). Plant Propagation. New Age International (P) Limited Publishers, New Delhi.

Semester-II

Course Title: UTILIZATION AND CONSERVATION OF PLANTS **Course Code: MDC-2**

Objectives: The basic objective of this paper is to make the students aware of origin of plants, their civilization, utilization and extinction and conservation.

Max. Marks : 75 **Course Credits =** 3
Theory : 75 Written Exam: 60 Internal Assessment: 15

Periods per week: Theory – Four of 45 minutes duration (Three Hours)

Instructions for Paper Setters:

Question paper will have four sections. Examiner will set a total of nine questions comprising of one compulsory question consisting of MCQ and fill in the blanks covering the whole syllabus and two questions from each unit. Student will attempt five questions in all – one question from each unit along with one compulsory question.

Teaching Aids**:** Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to black/white board**.**

Unit – I
Origin of agriculture-time and places of origin, domestication of cultivated plants; Forests resources –sustainable utilization and development; Ethnobotanyand its significance.
Unit – II
General account, food and economic value of the plants as source of: Fibres (Cotton and Jute); Pulses (Mung, Chickpea); Spices and Condiments (Cumin, Fennel, Ginger and Turmeric); Vegetables and fruits (Potato, Tomato, Grapes and Mango)**.**
Unit – III
General account, parts used and uses of the plants as source of: Vegetable oils (Sunflower and Mustard); Pulp plants (bamboo); Medicinal plants (Ashwagandha, *Aloe vera*); brief account of resins and gums; psychoactive drugs (Hemp and Poppy).
Unit – IV
IUCN categories of plants; plant extinction – causes and preventive measures; principles of conservation; *in situ* and *ex situ* conservation strategies; Conventions on Biodiversity (CBD); International agreement to protect species and habitat; Brief account of National Parks, Biosphere Reserves and Sanctuaries.

Suggested Readings:

1. Akeroyd, J. and Jackson, P.W. (1995). A Handbook of Botanic Garden and Reintroduction of Plants to the Wild. Botanic garden conservation Union, UK.
2. Chowdhery, H.J. and Murty, S.K. (2000). Plant Diversity and Conservation in India – an overview. Bishen Singh Mahendra Pal Singh, Dehradun.
3. Directory of Indian Wetlands (1993). W.W.F. India, New Delhi and AWB, Kuala Lumpur.
4. Heywood, V. (1995). Global Biodiversity Assessment. United National Environment Programme. Cambridge University Press, Cambridge, U.K.
5. Haunter, M.L. and Gibbs, J. (2007). Fundamentals of Conservation Biology. 3rd Edn. Blackwell Publishing, U.K.
6. Kothari, A. (1997). Understanding Biodiversity: Life Sustainability and Equity. Orient Longman.
7. Meffe, G.K. and Ronald, C.R. (1994). Principles of Conservation Biology. Sinauer Associates. INC Publishers, USA.
8. Primack, R.E. (2006). Essentials of Conservation Biology. 4th Edn. Sinauer Associates, U.S.A.
9. Rodgers, N.A. and Panwar, H.S. (1988). Planning a Wildlife Protected Area Network in India. Vol. I. The Report Wildlife Institute of India, Dehradun.

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Chemistry

Semester-I
IDC-1
Basic Analytical Techniques-I

Max. Marks: 75 (50T + 25P)
Total Credits: 03
Credits = 2+1 (T = 2; P = 1)

THEORY

UNIT-1

I) **Sampling, Collection, and Preservation of Samples**
Sampling and theories of sampling, Sampling procedure, Collection of Samples and its storage, Preparation of samples for analysis, calibration standards, Hazards in sampling, simple equilibrium calculations, General methods of preservation.

II) **Evaluation of Analytical Data (5 Lectures)**
Mean and standard deviation, Classification of errors, Minimization of errors, Accuracy and Precision, Distribution of random errors, errors in addition and multiplication, standard deviations, rejection of measurements, numerical problems related to the evaluation of analytical data.

III) **Volumetric and Gravimetric Methods of Analysis**
Theory of volumetric and gravimetric methods of analysis, preparation of solutions, equivalence point, standard solutions, Primary and Secondary standards, and point detection, theory of indicators and their selection for volumetric analysis, limitations, interferences.

UNIT-2

1. **Physicochemical Methods of Analysis**
a) Introduction, principle, and analytical applications of Conductometry, pH metry, and colorimetry. Acid-Base Titrations, Preparation of standard solutions of acids and bases, titration curves.

b) **Oxidation-Reduction Titration**,
Principle, Redox indicators, and applications of oxidation-reduction titrations.

2. Water Quality Analysis (8 Lectures)
a) Pure water, sources responsible for contaminating water, Hardness of water, purification methods, Determination of pH, acidity, and alkalinity of a water sample.

b) Complexation, formation constants, EDTA equilibria, effect of pH on EDTA equilibria, complexometric titration curves. Use of indicators, Applications of complexometric titrations, masking, and its utility in complexometric titration.

Suggested Reading
a. Svehla, G. Vogel's Qualitative Inorganic Analysis, Pearson Education, 2012.
b. Mendham, J. Vogel's Quantitative Chemical Analysis, Pearson, 2009.
c. Standard methods of Chemical Analysis, Welcher.

Laboratory Practical

Max. Marks: 25
Total Credits: 01

Conductometry:

1. Acid-Base titration.
2. Effect of concentration on equivalent conductance of a given electrolyte.

pH metry:

1. Acid-Base titration.

Redox Titrations:

1. Estimation of oxalic acid by titrating it with $KMnO_4$.
2. Estimation of water of crystallization in Mohr's salt by titrating with $KMnO_4$.

Gravimetric Analysis:

1. Determination of Nickel and Copper.

Complexometric Titrations:

1. Determination of hardness of water by the EDTA Method.

Suggested Reading:

a. Svehla, G. Vogel's Qualitative Inorganic Analysis, Pearson Education, 2012.
b. Mendham, J. Vogel's Quantitative Chemical Analysis, Pearson, 2009.
c. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J., & Smith, P.W.G., Textbook of Practical Organic Chemistry, Prentice-Hall, 5th edition, 1996.

Semester-II

IDC-2
Basic Analytical Techniques-II

Max. Marks: 75 (50T + 25P)
Total Credits: 03
Credits = 2+1 (T = 2; P = 1)

THEORY

UNIT-1

I) **Chromatography – 1 (5 Lectures)**
Introduction, principle, and classification of different chromatographic methods, methods of development – Elution development, Gradient elution development, displacement development, and frontal analysis. Dynamics of chromatography – efficiency of chromatographic column, zone spreading, High Equivalent Theoretical Plate (HETP), Van Deemter equation, resolution, qualitative and quantitative analysis.

II) **Chromatography – 2 (10 Lectures)**

- **Paper chromatography**: Principle, papers as a chromatographic medium, modified papers, solvent systems, mechanism of paper chromatography, experimental technique, different development methods – ascending, descending, horizontal, circular spreading, multiple development, visualization, and evaluation of chromatograms, applications.
- **Thin layer chromatography**: Principle, chromatographic media, coating materials, applications, activation of adsorbent, sample development, solvent systems, development of chromatoplate, visualization methods, and applications in separation.
- **Column Chromatography**: Principle, columns, matrix materials, stationary phase, column packing, application of sample, column development, and sample elution, detectors and fraction collectors, applications.

UNIT-2

I) **Advanced Technique I (8 Lectures)**
Gas Chromatography: Fundamentals and Principles of Gas Chromatography (GC), Instrumentation, Sample preparation, Carrier gases, Injectors, and Capillary Columns. Applications to separate, identify, and quantify samples like Petroleum products (Kerosene, Diesel, Petrol, Low boiling solvents, etc., and their residues).

II) **Advanced Technique II (7 Lectures)**
High Performance Liquid Chromatography: Principle and Instrumentation, Types of HPLC – Normal phase HPLC, Reverse Phase HPLC, column, matrices and stationary phases, mobile phase and pumps, detectors, applications on separation, identification, and quantification of amino acids.

Laboratory Practical

Max. Marks: 25
Total Credits: 01

1. Thin Layer Chromatography:

1. Determination of Rf values and identification of organic compounds. Separation of isomeric mixture of Ortho and para-nitroaniline using hexane and ethyl acetate (8.5: 1.5) by thin layer chromatography.
2. Separation of benzaldehyde and nitrobenzene using $KMnO_4$ solution as a visualizing reagent.

2. Paper Chromatography:

1. Separation of amino acids.
2. Separation of carbohydrate molecules.

3. Column Chromatography:

1. Separation of fluorescein and methylene blue.
2. Separation of leaf pigments from spinach leaves.

Suggested Reading:

a. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J., & Smith, P.W.G., *Textbook of Practical Organic Chemistry*, Prentice-Hall, 5th edition, 1996.
b. Mendham, J., *A.I. Vogel's Quantitative Chemical Analysis*, 6th Ed., Pearson, 2009.

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Computer Science and Application

Semester-I

Title:E-Commerce
Paper Code: BCA - 104

L	T	P	Cr	
3	-	-	3	Theory External Marks: 75
				Theory Internal Marks: --

Time Duration: 3 Hrs. Number of Theory Hours: 45

Objectives: The objective of this course is to understand the process of electronic commerce and familiarizes students with the technology involved in it.

Course Outcomes:
After completing the course students will be able to

i) Know about the Basic of E-commerce, its type, EDI, its structure and components
ii) Understand the basic of Web Based E-Commerce, Web Site Development and online promotion tools and techniques.
iii) Gain an insight into electronic payment systems, mobile commerce and various application of e-commerce along with some case studies.

Note :

i. The Question Paper will consist of Four Sections.
ii. Examiner will set total of **NINE** questions comprising **TWO** questions from each Section and **ONE** compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt **ONE** question from each Section and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT - I

An Overview of E-Commerce:
Definitions: E-commerce, E-business, difference between E-commerce and E-business, Problems with Traditional business systems, Aims of E-commerce, Types of E-commerce: B2B, B2C, C2C, B2G, G2H, G2C, Operational & Strategic benefits of E-commerce, Issues & Challenges in E-commerce.
Electronic Data Interchange (EDI): Definition; Traditional versus EDI enabled system for document exchange; Components of EDI: EDI Standards, EDI Software, Communication Networks; EDI Message Structure; EDI Notification Structure; EDI in India; EDI enabled procurement process; Benefits of EDI: Direct Benefits, Strategic Benefits; EDI Implementation issues; Legal Aspects

UNIT - II

Web based E-Commerce: Definition; Need for web based business, Steps in setting up business on Internet: Selection & registration of domain name, Website development: Planning a website, Steps for creating a website, Elements of a webpage, web authoring tools, Hosting a website: Website hosting considerations.
Online Promotion tools & techniques: Getting links to your site, banner advertisements & measuring advertisement effectiveness; Web Traffic Analysis: Hits, View pages, Visits and Other web-reporting tools, various measures, What is Search Engine optimization

UNIT - III

Electronic Payment Systems: E-cash: Purchasing & using of e-cash; Electronic Purses their loading with cash and use; E-cheque payment system; Online Third Party Verified Payment System through Credit & Debit Cards; ATM based cash disbursement system; Electronic Bill Payment System; Inter bank clearing system.

UNIT - IV

Mobile Commerce: Definition, Benefits of Mobile Commerce, Issues in Mobile Commerce, Mobile Commerce Framework

Applications of E-Commerce & Case Studies: Applications of e-commerce, Case studies in Retailing, Banking and e-governance; Cyber Crimes: Types, Cyber Forensics, Cyber crimes and IT Act - 2000.

Suggested Readings:

1. Bhasker, Bharat: Electronic Commerce: Framework, Technologies and Applications; Tata McGraw- Hill.
2. Bajaj, Kamlesh& Nag, Debjani: E-Commerce-The Cutting Edge of Business; Tata McGraw Hill.
3. Young, Margaret Levine : The Complete Reference: Internet; Tata McGraw Hill.
4. KalaKota, Ravi &Whinston, Andrew B.: Frontiers of Electronic Commerce; Addison Wesley.
5. Stallings, William: Network Security Essentials: Applications & Standards; Pearson Education.
6. Minoli, Daniel &Minoli, Emma: Web Commerce Technology Handbook; Tata McGraw Hill.
7. Murthy CSV: e-Commerce: Concepts, Models, Strategies; Himalayas Publishing House.
8. Kosiur, David: Understanding E-Commerce; Microsoft Press.

Semester-II
Management Information System
Paper Code: BCA-204

L	T	P	Cr	
3	-	-	3	Theory External Marks: 75
				Theory Internal Marks: --

Time Duration: 3 Hrs. Number of Theory Hours: 45

Objectives:

1. To understand and asses the importance of information and its role in business.
2. To develop data analyzing skills in students to evaluate information and the tools used for information processing.
3. To imbibe theoretical knowledge of MIS in the students and prepare the students technological competitive and make them ready to self-upgrade with the higher technical skill.

Course Outcomes:

1. Understand the information needs of an organization and a business function.
2. Evaluate effectiveness of decision making process and identify its tools
3. Understand DSS techniques for making effective decisions
4. Design parameters for MIS application, for data analysis uses

Note: Examiner will be required to set nine questions in all. First question will be compulsory, consisting of objective type/short answer type questions covering the entire syllabus. In addition to that eight more questions will be set, two questions from each unit. A candidate will be required to answer five questions in all, selecting one question from each unit in addition to compulsory question number one. All questions will carry equal marks.

UNIT-I

Introduction to System and Basic System Concepts, Types of Systems, The Systems Approach, Information System: Definition & Characteristics, Types of information, Role of Information in Decision-Making.

UNIT-II

An overview of Management Information System: Definition & Characteristics, Components of MIS, Frame Work for Understanding MIS: Information requirements & Levels of Management, Simon's Model of decision-Making, Structured Vs Unstructured decisions, Formal vs. Informal Systems.

UNIT-III

Developing Information Systems: Analysis & Design of Information Systems, Implementation & Evaluation, Pitfalls in MIS Development.

UNIT-IV

Functional MIS: A Study of Personnel, Financial and Production MIS, Introduction to EBusiness Systems, E-Commerce- Technologies, Applications, Decision Support SystemsSupport Systems for Planning, Control and Decision- Making.

TEXT AND REFERENCE BOOKS:

1. J. Kanter, "Management/Information Systems", PHI
2. Gordon B. Davis, M. H. Olson, "Management Information Systems- Conceptual Foundations, Structure and Development", McGraw Hill.
3. James A. O'brien, "Management Information Systems", Tata McGraw Hill.
4. Lucas, "Analysis, Design & Implementation", McGraw Hill
5. James A. Senn, " Analysis, Design of Information Systems", Second Edition, McGraw Hill.

(Any Semester)

Web Development/Designing

Course Objectives:

1. Identify the appropriate programming environment for developing dynamic client-side and server-side web applications.
2. Plan, develop, debug, and implement interactive client-side and server-side web applications.
3. Identify the tools needed to create dynamic client-side and server-side web applications.

Course Outcome:

1. Students are able to develop a dynamic web page by the use of JavaScript and HTML.

Theory External Marks-65

Theory Internal Marks-10

Credits- 3

Contact hours- 45

Time Duration: 3 hours

Note:

i. The Question Paper will consist of Four Sections.
ii. Examiner will set total of NINE questions comprising TWO questions from each Section and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Section and the Compulsory question.
iv. All questions carry equal marks unless specified.

Unit- I

World Wide Web – Web Browsers - Markup Languages - Style Sheet Technologies - Client-Side, Server Side – HTML – Headings – Links-Images – Lists - Tables - Forms, Frames, Cascading style Sheets

Unit - II

JAVASCRIPT- Introduction - Simple Program, Decision Making.
Equality and Relational Operators - Control Statements - Functions - Programmer Defined Functions, JavaScript Global Functions, Recursion

Unit - III

Arrays- References and Reference Parameters, Passing Arrays to Functions, Multidimensional Array Objects of JavaScript: Window Object - Document Object

Unit - IV

Forms Object -Text Boxes and Text Areas - Buttons, Radio Buttons, Check Boxes, Select Object, Other Object: Data Object- Math Object- String Object- Arrays- Worked Examples.

Practical Syllabus:

Create Webpages in HTML Links, Images, Lists, Tables, Forms, Frames, Cascading style sheets
Creating Webpages in JavaScript, Simple Program, Decision Making, Equality and Relational Operators - Control Statements-Functions-Programmer Defined Functions, JavaScript Global Functions, Recursion-Arrays

Create web pages in JavaScript-Window Object-Document object-Forms- Object, Text Boxes and Text Areas, Buttons, Radio Buttons, Check Boxes, Select Object, Other Object, Data Object, Math Object, String Object, Arrays, Worked Examples

References:

1. Learning Mike McGrath, JavaScript, Dream, Tech Press, 1st Edition 2006
2. Jennifer Robinson, Learning Web Design
3. Melani and Kyrin's "HTML, CSS, JavaScript All in One"
4. HTMLS Black Book by DT Editorial Services

(Any Semester)

Introduction to Information Technology

Course title	Credits	Credit distribution of the course			Pre-requisite of the course (if any)
		Lecture	Tutorial	Practical/ Practice	
Introduction to Information Technology	3	0	0	3	NIL

Objectives:

1. To familiarize the students with the Fundamentals of Information Technology.
2. To assess the implications for the markets and organizational change of advances in information technologies.

Course Outcomes:

- Knowledge about the basics of Windows Operating System.
- Learning various operations of Windows OS.
- Knowledge of MSword, MS Power Point, MS Excel, MS Access

UNIT I

Introduction:Evolution of Computers, Windows and Unix. Fundamentals of windows, anatomy of windows, Operations on window: Opening, Minimizing, Maximizing, Moving, Resizing, Closing; Windows Explorer, Folders: Creating, deleting, copying, renaming folders, folder properties; Icons, Menu, Taskbar, Control panel, Recycle bin.

UNIT II

Word Processing Package: Basics of Word Processing; Word Processing Basics; Text creation, Manipulation, Finding and replacing, Formatting of text; Printing of word document, Page Layout: Margin setting, Alignments, Adding Borders and shading, Adding Headers and Footers, Setting up Multiple columns, Working with tables, Spell check, Auto Correct, Grammar facility, Mail merge.

UNIT III

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of numbers, text and formulae, Moving data in a worksheet, Moving around in a worksheet, Selecting Data Range. Using the interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing: Absolute, Relative and Mixed; Formatting and Calculations: usingAutofill, Working with Formulae, Creating Chart and graphs.

UNIT IV

Presentation Packages: Basics, General Features, Creating a presentation, Different types of slide views, Master Slides and its use, Formatting Slides: slide design, Layout and background; Animation effect, Transition effect, timing effects, Macros.

Database Package: Introduction to Database, Tables, Data Types, Attributes, Records; Overview of MS-ACCESS Creating Database, Creating Tables, Data types, Creating forms and queries.

References Resources:

1. Sinha, P.K.: Computer Fundamentals
2. Rajaraman, V.: Fundamentals of Computers.
3. Texali, R.K.: PC Software made simple, Tata McGraw Hill.

Fashion Designing

SEMESTER-I

INTRODUCTION TO FAMILY AND COMMUNITY SCIENCE (THEORY)

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2 hours per week **Paper: 40**
Duration of Exam: 3 hours **Internal Assessment: 10**

Instructions to the Examiner

- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Course Outcome:

Students will have an in depth understanding of the following:

- Understanding of the self and one's role and responsibilities as a productive individual and as a member of family, community and society.
- Discipline of Family and Community Science for professional careers.
- Knowledge specific to five domains namely, Foods and nutrition, Human Development, Clothing and Textiles, Family Resource Management and Extension Education.
- Functional skills in the five domains for career and employment

UNIT-I

1. Introduction & Definition of Family and Community Science
   - Core areas of Family and Community Science
   - Scope & opportunity in Family and Community Science

2. Development of Family and Community Science as a discipline

UNIT II

3. Foods & Nutrition
   - Basic terminology- Food, Nutrition, Food groups, Balance Diet, Food preservation, Meal planning, Nutritional deficiencies and Food hygiene
   - Role & Importance of Foods & Nutrition in Community
   - Functions of food
   - Basic Food groups: sources and their functions

4. Human Development
   - Basic terminology- Growth, Development, Childhood, Adolescent, Anxiety, Expression, Fear, Personality, Sex Education
   - Role & Importance of Human Development in Community
   - Stages of growth and development

UNIT III

5. Clothing & Textiles
   - Basic terminology- Fiber, Yarn, Spinning, Fabric, Apparel, Textiles, Laundry, Starch, Blues, Weaving, Knitting, Non-wovens
   - Role & Importance of Clothing & Textiles in Community
   - Care and maintenance of garments.

6. Resource Management
   - Basic terminology- Home Management, Human resources, Time management, Work simplification, Ergonomics, Work ethics, Family Budget, Furniture & Furnishings, Floor coverings
   - Role & Importance of Resource Management in Community
   - Time, energy and money management

UNIT IV

7. Extension & Communication
   - Concept, Nature & Scope of Extension
   - Family and Community Science Extension as a Discipline and its contribution towards development
   - Methods & Media of Community Outreach
   - Role & Importance of Extension & Communication in Community and National Development.

Recommended Readings

1. Basic Food Preparation: A Complete Manual, Faculty of Lady Irwin College, Orient Longman publishers.
2. Bhatnagar. O.P &Dhama, O.P. (2009). Education And Communication For Development 2ed. New Delhi: Oxford & IBH Publishing Co. Pvt Ltd. ISBN-13: 978-8120400306.
3. Dantyagi S, "Fundamentals of Textiles and their Care," Orient Longman Ltd, New Delhi.
4. Goyal, M. (2022). Home Science. SBPD Publishers
5. R Vatsala "Textbook of Textiles & Clothing" (2003), ICAR, New Delhi
6. Seetharaman P, Batra S. and Mehra P 2005. An Introduction to Family Resource Management. CBS Publishers and distributors, New Delhi.
7. Sumati R Mudambi & MV Rajagopal (2004). Fundamentals of Foods & Nutrition. Forth edition, New Age P Publishers.

SEMESTER-I

INTRODUCTION TO FAMILY AND COMMUNITY SCIENCE (PRACTICAL)

Credit: 1 **Maximum Marks: 25**
Duration of class: 2 hours per week **Paper: 20**
Duration of Exam: 3 hours **Internal Assessment: 05**

Instructions to examiner

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Course Outcome:

Students will have an in depth understanding of the following:

- Develop an understanding of the self and one's role and responsibilities as a productive individual and as a member of family, community and society.
- Learning across diverse domains and undertake a critical analysis of issues and concerns specific to family, community and society.
- Learners with the basic knowledge specific to five domains namely, Foods and nutrition, Human Development and Family studies, Fabric and Apparel, Resource Management and Communication and Extension.

Content

1. Preparation of Audio–Visual Aids on following topics-
    - Food Pyramid
    - Menu Planning for different occasions
    - Teaching aids for pre-school children
    - Table setting
    - Textile labels
2. Planning & Arrangement of various thematic parties (Decoration, Table setting, Meal planning and Fashion styling)

Note- Marks will be awarded by External on the basis of sessional work presentation

Recommended Readings

1. Sumati R Mudambi& MV Rajagopal (2004). Fundamentals of Foods & Nutrition. Forth edition, New Age P Publishers.
2. Basic Food Preparation: A Complete Manual, Faculty of Lady Irwin College, Orient Longman publishers, 2008.
3. Seetharaman P, Batra S. and Mehra P 2005. An Introduction to Family Resource Management. CBS Publishers and distributors, New Delhi.
4. Bhatnagar. O.P &Dhama, O.P. (2009). Education And Communication For Development 2ed. New Delhi: Oxford & IBH Publishing Co. Pvt Ltd. ISBN-13: 978-8120400306.
5. Goyal, M. (2022). Home Science. SBPD Publishers
6. Dantyagi S, "Fundamentals of Textiles and their Care," Orient Longman Ltd, New Delhi.
7. R Vatsala "Textbook of Textiles & Clothing" (2003), ICAR, New Delhi

SEMESTER –II

SUSTAINABILITY IN EVERYDAY LIFE
THEORY

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2 hours per week **Paper: 40**
Duration of Exam: 3hours **Internal Assessment: 10**

Instructions to the Examiner

- Question paper will have four sections. Paper setter will set a total of nine questions comprising of one question from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Course Outcome:
Students will have an in depth understanding of the following:

- Terms related to sustainability
- Sustainable practices at home

Content:

UNIT –I

1. Terminology: Sustainability, 3R's of sustainability, Eco-labels, Biodegradable and Non Biodegradable waste, E-Waste , Zero waste concept, Carbon Foot Print, Fair trade, Green washing, Life cycle assessment, Circular economy

UNIT –II

2. Sustainable Development: Definition and Concept
3. Five P's of Sustainable Development
   People, planet, prosperity, peace and partnership
4. Factors of Sustainability

UNIT –III

5. Sustainable practices at home
6. Different ways to be sustainable in everyday life
7. Sustainable brands in everyday life

UNIT –IV

8. Waste management: Definition and Types
9. Concept of Waste management
10. Government Initiatives for Sustainable development

Recommended readings:

1. Cradle to Cradle: Remaking the Way We Make Things By William McDonough & Michael Braungart
2. The Upcycle: Beyond Sustainability – Designing for Abundance by William McDonough & Michael Braungart
3. Grow Food For Free: The sustainable, zero-cost, low-effort way to a bountiful harvest by Huw Richards
4. Strategy for Sustainability: A Business Manifesto by Adam Werbach
5. Zero Waste Sewing: 16 projects to make, wear and enjoy by Elizabeth M Haywood Cooatalaa Press
6. 50 Nifty Thrifty Upcycled Fashions: Sew Something from Nothing by Cynthia Anderson, Stackpole Books
7. Wear, Repair, Repurpose: A Maker's Guide to Mending and Upcycling Clothes by Lily Fulop, Countryman Press
8. Recycling in Textiles by YoujiangWang , Woodhead Publishing Ltd
9. Wise Craft Quilts: A Guide to Turning Beloved Fabrics Into Meaningful Patchwork by Blair Stocker, Roost Books
10. Creative Recycling in Embroidery by Val Holmes , Batsford Ltd
11. Waste Recycling and Reuse by Sterling , Sterling Publishers
12. Reduce, Reuse, Recycle: An Easy Household Guide (Green Books Guides) by Nicky Scott (Author), Axel Scheffler , Green Books
13. The Complete Technology Book on E-Waste Recycling ,NPCS Board of Consultants & Engineers, Asia Pacific Business Press Inc.
14. Household Waste Recycling by Dr. Richard Waite , Earthscan Ltd

SEMESTER –II

SUSTAINABILITY IN EVERYDAY LIFE
PRACTICAL

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2 hours per week **Paper: 20**
Duration of Exam: 3 hours **Internal Assessment: 05**

Instructions to examiner:

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Course Outcome
Students will have an in depth understanding of the following:

- Understand Terms related to sustainability
- Sustainable practices at home

Content:

1. Preparation of Audio- visual aids on sustainability and sustainable practices.
2. Awareness generation on sustainability and sustainability practices through Prepared audio-visual aids in neighbourhood (On Campus and off Campus).
3. Product development from waste (5 different products will be developed from different mediums).
4. Preparation of file and presentation on field work.

Recommended readings:

1. Cradle to Cradle: Remaking the Way We Make Things By William McDonough & Michael Braungart
2. The Upcycle: Beyond Sustainability – Designing for Abundance by William McDonough & Michael Braungart
3. Grow Food For Free: The sustainable, zero-cost, low-effort way to a bountiful harvest by Huw Richards
4. Strategy for Sustainability: A Business Manifesto by Adam Werbach
5. Zero Waste Sewing: 16 projects to make, wear and enjoy by Elizabeth M Haywood Cooatalaa Press
6. 50 Nifty Thrifty Upcycled Fashions: Sew Something from Nothing by Cynthia Anderson, Stackpole Books
7. Wear, Repair, Repurpose: A Maker's Guide to Mending and Upcycling Clothes by Lily Fulop, Countryman Press
8. Recycling in Textiles by YoujiangWang , Woodhead Publishing Ltd
9. Wise Craft Quilts: A Guide to Turning Beloved Fabrics Into Meaningful Patchwork by Blair Stocker, Roost Books
10. Creative Recycling in Embroidery by Val Holmes , Batsford Ltd
11. Waste Recycling and Reuse by Sterling , Sterling Publishers
12. Reduce, Reuse, Recycle: An Easy Household Guide (Green Books Guides) by Nicky Scott (Author), Axel Scheffler , Green Books
13. The Complete Technology Book on E-Waste Recycling ,NPCS Board of Consultants & Engineers, Asia Pacific Business Press Inc.
14. Household Waste Recycling by Dr. Richard Waite , Earthscan Ltd

Home Science

Semester-I
Introductory Life Sciences for Home Science

Theory

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2 hours per week **Paper: 40**
Duration of Exam: 3 hours **Internal Assessment: 10**

Instructions to the Examiner

- Each theory paper will be of three hours duration.
- Question paper will have four sections. Paper setter will set a total of nine questions comprising of one question from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/unit including the compulsory question.
- All questions may carry equal marks, unless specified.

Course Outcome

- The student will be able to understand knowledge of Life Science in day to day life

Applied Botany

Unit-I

1. Study of Vegetative Propagation by artificial Methods- cutting, layering, budding, and grafting.
2. Principle and planning in layout of kitchen Garden.

Unit-II

3. Principle and planning in layout of a Garden.
4. Organic manures, Inorganic Fertilizers, Vermi compost and Bio fertilizer

Applied Zoology

Unit-III

5. Types, Structure and function of animal cells and its components
6. Identification of stored grain pest *Sytophillus, Rizopertha, Callosobruchus*and*Tribolium*
7. Transmission, prevension and control of Taeniasis, Ascariasis, Malaria and COVID-19, Vermi composting

Unit-IV

8. An Elementary Knowledge of Structure and Function of DNA and RNA.
9. Heredity- Simple Inheritance according to Mandel-law, Monohybrid and dihybrid cross, Incomplete inheritance, an Elementary Knowledge of Autosomal and sex chromosomal abnormalities, an elementary knowledge of ABO Blood Grouping and its genetic basis, elementary knowledge of Genetic basis Haemophilia, Colour-blindness, Mongolism

Recommended Readings

1. NCERT Biology BOOK class 11 & 12
2. Dhami: P.S, A text book of Zoology
3. David Hames & Nigel Hooper: Instant Notes Biochemistry

Semester- I

Introductory Life Sciences for Home Science

Practical

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2 hours/week **Paper: 20**
Duration of Exam: 3 hours **Internal Assessment: 05**

Instructions to Examiner

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Course Outcome

- The student will be able to apply knowledge of life science in day to day life

Content

Applied Botany

1. Preparation of a pot for sowing seeds and study different seed sowing methods
2. Preparation of bed for transplanting vegetables, potato sowing and raising seedlings
3. Demonstration of methods of vegetative propagation by cutting and grafting methods
4. To prepare temporary slide on onion peel

Applied Zoology

5. Identification of few Common animal (Insect pest, Fish etc.), Birds in human environment
6. Determination of ABO Blood group
7. Demonstration of Vermi composting

Recommended Readings

1. NCERT Book , Class 11 & 12
2. Home Gardening- Shashi Kumar
3. A test Book of Practical Zoology, Vertebrate, by S.S.Lal.

Semester-II

Introductory Physical Sciences for Home Science

Theory

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2 hours/week **Paper: 40**
Duration of Exam: 3 hours **Internal Assessment: 10**

Instructions to Examiner

- Each theory paper will be of three hours duration.
- Question paper will have four sections. Paper setter will set a total of nine questions comprising of one question from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/ units including the compulsory question.
- All questions may carry equal marks, unless specified.

Course Outcome

- The student will be able to understand knowledge of physical science in day to day life

Applied Physics

Unit – I

1. Principle, construction, working and applications of Optical Fibres.
2. Applications of Lasers – Medical, Scientific, Industry and Commercial.
3. Safety features in household electric wiring- Fuse, MCB, Earthing.

Unit – II

4. Applications/illustrations of Conduction, Convection and Radiations in day to day life
5. Radioisotopes and their uses in Medicine, Industry, Agriculture
6. Radiation hazards and Safety Measures

Applied Chemistry

Unit – III

1. Essentials of Chemistry: Symbols formulae, Valency and Variable valency, elementary idea of Impirical formula and Molecular formula (no numerical) definition of Atomic Weight and Molecular Weight, Mole Concept.
2. Applications of Chemistry: Soaps & detergents; introduction and their Cleansing action. Cosmetics; brief study and elementary idea about ingredients- cold cream, vanishing cream, lipstick, mascara. Food Additives.

Unit – IV

3. Polymerization and Polymers- Definition and classification. Polymers in textiles: Chemistry of synthetic fibres - Nylon, Polyester and Acrylic fibers.
4. Fertilizers: Nitrogen, Potassium and Phosphorus

Recommended Readings

1. Avery M., Household Physics
2. Ajoy Ghatak, K. Thagarajan**,**Lasers: Introduction to Fiber Optics
3. Gomber & Gogia, Pardeep's Fundamental Physics (XI, XII)
4. Gupta S.K, Modren's ABC of Physics (XI, XII)
5. Lal S., Fundamental Physics (XI, XII)
6. Ajoy Ghatak, K. Thagarajan**,** Lasers: Fundamentals and Applications (Graduate Texts in Physics)
7. Sachdeva & Duggal, Intermediate Physics
8. Applied Chemistry for Home Science and Allied Science by Thancamma, Jacob

Semester II
Introductory Physical Sciences for Home Science

Practical

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2 hours per week **Paper: 20**
Duration of Exam: 3 hours **Internal Assessment: 05**

Instructions to Examiner

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Course Outcome

- The student will be able to apply knowledge of physical science in day to day life

Content

Applied Physics

1. Demonstration of Solar Cooker
2. Demonstration of Energy efficient light bulbs
3. Demonstration of conduction, convection, radiation with respect to cooking

Applied Chemistry

4. Preparation of vanishing cream and cold cream
5. Preparation of detergent and liquid soap
6. Determination of degree of hardness of tap water volumetrically and determination of melting point of organic compounds

Recommended Readings

1. Gupta S.K., Modern ABC of Physics
2. Jaiswal J. N., Comprehensive Practical Physics
3. Applied Chemistry for Home Science and Allied Science by Thancamma, Jacob

Semester-I

Basics of Nutrition and Food Science - I

THEORY

Credit Hours: 2 **Maximum Marks: 50**
Duration of class: 2 hours per week **Paper: 40**
Duration of exam: 3 hours **Internal Assessment: 10**

INSTRUCTIONS TO EXAMINERS:

1. Each theory paper will be of three hours duration.
2. The question paper will have four sections/units. Paper setter will set a total of nine questions, comprising of two questions from each section and one compulsory question of short-answer type covering the whole syllabus.
3. Students will attempt one question from each unit and the compulsory question (Total of five questions).
4. All questions may carry equal marks, unless specified.

OBJECTIVES:

1. To introduce the concept of different areas of Foods and Nutrition.
2. To gain knowledge about Foods and Nutrition.

UNIT I

1. Introduction to Foods and Nutrition
    - Definition of food
    - Functions of Food - physiological, psychological, and social
2. Classification of food on the basis of its functions
    - Health, Optimum Nutrition, Malnutrition
    - Basic terminology used in cooking (refer annexure)

UNIT II

3. Functions and sources of:
    - Macronutrients: Carbohydrates, Proteins, and Fats
    - Micronutrients: Minerals - Calcium, Iron, and Iodine; Vitamins: Fat-soluble (A, D, E, K) and Water-soluble (thiamine, riboflavin, niacin, pyridoxine, cyanocobalamin B12, ascorbic acid)
4. Functions and sources of:
    - Water
    - Fiber

UNIT III

5. Objectives of cooking food
6. Methods of cooking:
    - Dry heat, moist heat, cooking with oil
    - Solar cooking and microwave cooking
7. Effect of cooking on nutrients

UNIT IV

8. Meal planning
    - Fundamentals of meal planning
    - Factors affecting meal planning
9. Food safety
    - Basic concept of food adulteration
    - Introduction to food safety laws (PFA, BIS, HACCP, ISO, ISI, FPO, FSSAI)

ANNEXURE

(Basic terminology)

a) Alacarte
b) Appetizer
c) Augratin
d) Batter
e) Blanching
f) Braising
g) Croutons
h) Dust
i) Emulsion
J) Foldin
k) Garnish
l) Glaze
m) Grilling
n) Marinate
o)Meringue
p) Puree
q) Roasting
r) Sauté
s) Steam
t) Whip
u) Zest

SEMESTER-I

Basics of Nutrition and Food Science - I
Practical

Credit Hours: 1 **Maximum Marks: 25**
Duration of class: 2 hours per week **Paper: 20**
Duration of exam: 3 hours **Internal Assessment: 5**

OBJECTIVES:

1. To understand the concepts of weights and measurements (raw and cooked food) and its importance.
2. To acquire skills in food preparation techniques.
3. To use appropriate methods of cooking for preparation of specific food products.
4. To observe and understand the principles involved in the preparation of different foodstuffs.
5. To learn some methods of preservation of foods.
6. To understand the concept of food adulteration.

Syllabus:

1. Identification of foods.
2. Weights and measures.
3. Concept of portion cooking.
4. Raw and cooked weight of foods.
5. Developing and preparing recipes rich in proteins, carbohydrates, fat, fiber, calcium, iron, vitamin A, and vitamin C.
6. Detecting adulteration in foodstuffs – ghee, honey, coffee, milk, turmeric (haldi).

RECOMMENDED READINGS:

1. *Discovering Nutrition*, Paul Insel, Don Ross, Kimberley McMahon, Melisa Bernstein, Jones and Bartlett Learning, 4th Edition, 2012.
2. *Food Hygiene and Sanitation*, Sunetra Roday, Tata McGraw-Hill Education Private Limited, 2011.
3. *Food Science and Nutrition*, Sunetra Roday, Oxford University Press, 2008.
4. *Food Science*, B. Srilakshmi, New Age International Publishers, 2012.
5. *Foods Facts and Principles*, N. Shakuntala Manay, M. Shadaksharaswamy, New Age Publishers, 2013.
6. *Fundamentals of Meal Management*, Margaret McWilliams, Pearson, 2008.
7. *Textbook of Nutrition and Dietetics*, Kumud Khanna, Elite Publishing House Pvt. Ltd, 2008.
8. *Nutrition and Dietetics*, Shubhangini A. Joshi, Tata McGraw-Hill Publishing Company Limited, 2009.
9. *Basic Food Preparation: A Complete Manual*, Faculty of Lady Irwin College, Orient Longman Publishers, 2008.
10. *Textbook of Foods and Nutrition and Child Development*, Kaur J, Gupta S, Garg N, Aggarwal A, Gill P (2023), Kalyani Publishers.

Semester-II

Basics of Nutrition and Food Science - II
THEORY

Credit Hours: 2 **Maximum Marks: 50**
Duration of class: 2 hours per week **Paper: 40**
Duration of exam: 3 hours **Internal Assessment: 10**

INSTRUCTIONS TO EXAMINERS:

1. Each theory paper will be of three hours duration.
2. The question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each section and one compulsory question of short-answer type covering the whole syllabus.
3. Students will attempt one question from each unit and the compulsory question (Total of five questions).
4. All questions may carry equal marks, unless specified.

OBJECTIVES:

1. To introduce the concept of different areas of Foods and Nutrition.
2. To gain knowledge about Foods and Nutrition.

UNIT I

1. Study of the food groups with respect to their classification, basic composition, and nutritive content:
    - Cereals
    - Pulses and legumes
    - Fruits and vegetables
    - Fats and oils
    - Sugar and jaggery
2. Beverages: tea, coffee, cocoa

UNIT II

3. Study of the food groups with respect to their classification, basic composition, and nutritive content:
    - Milk and milk products
    - Eggs, meat, fish, and poultry
    - Nuts and oilseeds
4. Food storage:
    - Introduction and general guidelines for safe food storage
    - Dry food storage
    - Refrigerated store
    - Freezer storage

UNIT III

5. Balanced Diet:
    - Definition and concept
    - Factors affecting balanced diet
6. RDA (Recommended Dietary Allowances) – definition:
    - Reference man and woman
    - Food pyramid and food plate

UNIT IV

7. Food Preservation:
    - Definition and importance
    - Food spoilage: causes and factors affecting it
    - Principles of food preservation
8. Methods of preservation – household and commercial

SEMESTER-II

Basics of Nutrition and Food Science - II
Practical

Credit Hours: 1 **Maximum Marks: 25**
Duration of class: 2 hours per week **Paper: 20**
Duration of exam: 3 hours **Internal Assessment: 5**

OBJECTIVES:

1. To understand the concepts of weights and measurements (raw and cooked food) and its importance.
2. To acquire skills in food preparation techniques.
3. To use appropriate methods of cooking for preparation of specific food products.
4. To observe and understand the principles involved in the preparation of different foodstuffs.
5. To learn some methods of preservation of foods.
6. To understand the concept of food adulteration.

Syllabus:

1. Food preparation according to different food groups:
    - Cereals and cereal products
    - Pulses – husked, de-husked
    - Vegetables and fruits
    - Milk and milk products
2. Beverages preparation
3. Demonstration of preparation of jams, murrabba, pickles for preservation

RECOMMENDED READINGS:

1. *Discovering Nutrition*, Paul Insel, Don Ross, Kimberley McMahon, Melisa Bernstein, Jones and Bartlett Learning, 4th Edition, 2012.
2. *Food Hygiene and Sanitation*, Sunetra Roday, Tata McGraw-Hill Education Private Limited, 2011.
3. *Food Science and Nutrition*, Sunetra Roday, Oxford University Press, 2008.
4. *Food Science*, B. Srilakshmi, New Age International Publishers, 2012.
5. *Foods Facts and Principles*, N. Shakuntala Manay, M. Shadaksharaswamy, New Age Publishers, 2013.
6. *Fundamentals of Meal Management*, Margaret McWilliams, Pearson, 2008.
7. *Textbook of Nutrition and Dietetics*, Kumud Khanna, Elite Publishing House Pvt. Ltd, 2008.
8. *Nutrition and Dietetics*, Shubhangini A. Joshi, Tata McGraw-Hill Publishing Company Limited, 2009.
9. *Basic Food Preparation: A Complete Manual*, Faculty of Lady Irwin College, Orient Longman Publishers, 2008.

Semester-I
Basics of Fashion Designing-I

Scheme of Examination
Credits: 3

Theory

No. of Paper	Time in hrs	Marks Allotted	Int. Ass.	Total Marks
1	3	40	10	50

Practical

Sr. No.	Name of Paper	No. of papers	Time in hrs	Marks Allotted	Int. Ass.
1	Basics of Fashion Designing-I	1	3	20	5

Theory
Credits:2
Maximum Marks : 50
Duration of Class : 2 Hours per week
Paper :40
Duration of Exam: 3 Hours
Internal Assessment: 10

INSTRUCTIONS FOR THE PAPER SETTER:
The question paper will consist of 5 sections: A, B, C, D, and E.
Sections A, B, C, and D will have 2 questions from the respective sections of the syllabus and will carry 8 marks each.
Section E will consist of objective type questions covering the entire syllabus uniformly and will carry 8 marks.

INSTRUCTIONS FOR THE CANDIDATES:
Candidates are required to attempt one question each from the sections A, B, C, and D of the question paper and the entire section E.

Note:
1. Internal assessment should be based on Assignment/test/Seminars and Attendance for Theory and Practical.
2. Practical examination will be held before the final Theory Exams.
3. Practical groups should have 12-15 students.

Section A
1. Equipment and supplies used for clothing construction - their use and care
2. Sewing machine :(a). Parts of sewing machine and its accessories (b). Common defects in sewing machine and their remedies (c). Care for a sewing machine

Section B.

3. Terminology(Grain, Bias, Selvadge, Ease, Seam Allowance, Dart)
4. Recording of Body Measurements and Precautions to be taken while taking body measurements
5. Preparation of Fabric for Clothing Construction, Shrinking and Straightening
6. Methods of Pattern Development (Drafting)

Section C

2. Fashion Terminology: Apparel, Fashion, Fad, Craze, High Fashion, Mass Fashion, Style, Change, Classic, Boutique, Croquet, Silhouette, Designer, Collection, Adaptation
3. Fashion Trend, Fashion Cycle, Sources of Fashion, Factors Favoring Fashion, Selecting Fashion, Forcasting Fashion, Fashion Show, Advertising and Display

Section D

4. Elements of Art such as Color, Form, Texture, Shape and Line
5. Principles of Design such as Harmony, Balance, Rhythm, Emphasis, Proportion

BASICS OF FASHION DESIGNING –I (PRACTICAL)

Credits: 1
Maximum Marks: 25
Duration of Class: 2 Hours per week
Paper: 20
Duration of Exam: 3 Hours
Internal Assessment: 5

Note:-Paper will be set on the sport by the examiner.

1. Make samples of the following:
   (a). Tacking, Running Stitch, Hamming, Back Stitch, Fasteners using Buttonhole Stitch
   (b). Seams -Flat Seams, Counter Hem, Mantua Maker, French Seam & Run & Fell Seam
   (c). Processes- Continuous Wrap, Two-Piece Placket Opening, Pleats, Gathers into a Band, Tucks
   (d). Joining- Shaped Facing on V Neckline and Piping on Round Neck
2. Embroidery- any "10" fancy Stitches
3. Drafting of the following:
   a. Jangia
   b. Child's Bodice Block
   c. Sleeve, Paint Sleeve, Puff Sleeve
   d. Collars, Baby, Cape and Peter Pan
4. Construction of the following:
   a. Jangia
   b. Gathered Frog with any Sleeve and Collar

Semester-II

BASICS OF FASHION DESIGNING - II

Theory

No. of Paper	Time in hrs	Marks Allotted	Int. Ass.	Total Marks
1	3	40	10	50

Practical

Sr. No.	Name of Paper	No. of papers	Time in hrs	Marks Allotted	Int. Ass.
1	Basics of Fashion Designing-II	1	3	20	5

Theory
Credits: 2
Maximum Marks: 50
Duration of Class: 2 Hours per week
Paper: 40
Duration of Exam: 3 Hours
Internal Assessment: 10

INSTRUCTIONS FOR THE PAPER SETTER:
The question paper will consist of 5 sections: A, B, C, D, and E.
Sections A, B, C, and D will have 2 questions from the respective sections of the syllabus and will carry 8 marks each.
Section E will consist of objective type questions covering the entire syllabus uniformly and will carry 8 marks.

INSTRUCTIONS FOR THE CANDIDATES:
Candidates are required to attempt one question each from the sections A, B, C, and D of the question paper and the entire section E.

Note:
1. Internal assessment should be based on Assignment/test/Seminars and Attendance for Theory and Practical.
2. Practical examination will be held before the final Theory Exams.
3. Practical groups should have 12-15 students.

Section A

1. Selection of suitable clothes for following age groups:
   Infants, Toddlers, School-going children, Adolescents, Adults, Elderly
2. Methods of laundry/washing
3. Care and storage of garments of cotton, wool, silk

Section B

4. Bleaches- oxidizing, reducing bleaches and their suitability to different fabrics
5. Blues-types and uses

Section C

6. Application of color on fabrics
    (a). Dyeing, simple dyeing of cotton
    (b). Resist- Tie, Dye and Batic
7. Printing-
   Block printing, screen printing, roller printing

Section D

8. Traditional Embroideries of India:
    1. Kantha
    2. Phulkari
    3. Chikankari
    4. Chamba
    5. Kasuti
    6. Kutch

BASICS OF FASHION DESIGNING - II (PRACTICAL)

Credits: 1
Maximum Marks: 25
Duration of Class: 2 Hours per week
Paper: 20
Duration of Exam: 3 Hours
Internal Assessment: 5

Note:-Paper will be set on the sport by the examiner.

Syllabus

1. Preparation of samples of Tie and dye & block printing
2. Drafting and construction of the following garments:
   a. Petticoat
   b. Lady's Kameez and Salwar
3. Project on care labels of garments or market survey of laundry products
4. Samples of Traditional Embroideries:. Kutch, Phulkari, Kashida, Chamba, Kantha, Chikankari, Kasutiand. Sindhi
5. Sample of applique work

REFERENCES:

1. Singer Sewing Machine, Mary Brooks, Bichen, McGraw, Hill Book Company, Inc, New York, Toronto, London, 1953.
2. Textile Fabrics and their Selection, Isabel B. Wingate, Prentice Hall, INC, Englewood, Clifton, NJ, 1970.
3. Fundamentals of Textiles and their Care, SushilaDantyagi, Orient, Longsmans, Bombay, Kolkata, Madras, New Delhi, Reprinted, 1968.
4. Household Textiles and Laundry Work, DurgaDeulkar, Atma Ramand Sons, New Delhi, 1973.
5. Ideas for Fabric Printing and Dying Gooch, Peter H. Charles, Cribner's, New York, 1974.
6. Encyclopedia of Textiles, Fabrics, Grayson Martin, John Wiley, New York, 1984.
7. Tie-Dyed Textiles of India-Veronica Murphy and Rosemary Crill, Pub. Mapiu Publishing Ltd, Ahmedabad.
8. Fashion and Textile, Neelima
9. Printing and Washing of Textiles, Neelima
10. Handbook of Fashion and Textile Designing, R.K. Sahu
11. Traditional Embroidery of India, Shailjan D. Naik
12. Basic Processes and Clothing Construction, SherieDoongaji&RaushniDeshpande, RaajPrakashan, New Delhi (4th Revised Edition)
13. Gupta L, Gupta S, Garg N, Kaur J & Gil P., "Textbook of Clothing, Textiles, Laundry and Fashion Designing",Kalyani Publishers, New Delhi 2022
14. Patron Drafting Vol. III, Dress Making, Kamakura-Shobho Publishing, Co. Ltd., Tokyo, Japan, 1972 (5th Edition)
15. A Manual of Children Clothing,SavitriPandit, Orient Longmans Ltd., Bombay, 1967
16. Practical Dress Design, Principles of Fitting and Patron Making, Mabel D. Erwin, The Macmillan Company, New York, 9th Printing, 1966
17. Indian Embroidery, Its Variegated Charms, SavitriPandit, Faculty of Home Sciences, Baroda, 1976, 1st Edition
18. The Basic Book of Sewing, Eve Harlow Pub, Octo Pits Book Ltd., London
19. Simple Dressmaking II, Maureen Gudd, Sworthy Pub. Milk & Boon, Ltd., London
20. Encyclopaedia of Dress Making, Pub Marshall Caven Dish Books Ltd., London

Mathematics

SEMESTER-I
MDC-MAT-101: BASIC MATHEMATICS – I

Credits = 3 (L = 2 + T = 1)
1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.
So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. / week
Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)
Exam Time: 3 hours **Max. Marks:** 75
(68 External + 7 Internal Assessment)

Note:

1. The syllabus has been split into three Units: Unit – I, Unit – II, and Unit – III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective:
The objective of this course is to study the basics of various topics of Mathematics which is a foundation for further learning in mathematics, physics, statistics, etc.

Learning outcomes:
The course will enable the students to learn the following:

1. Application of set theory and permutation combination.
2. Basics of matrices and their determinant.
3. Basic statistics, mean deviation, variance, and standard deviation.
4. Graphical representation of the solution of linear inequalities and its application.

Unit – I

Sets, Relations, and Functions, Principle of Mathematical Induction, Permutation and Combinations {Scope as in chapters 1, 4 of (A)}. Matrix, type of Matrices, operations on matrices, Transpose of Matrix, Symmetric and Skew Symmetric Matrices. {Scope as in chapters 3 (3.1 to 3.6) of (B) Part-I}

Unit – II

Determinant, Properties of Determinants, Minor and Cofactors, Adjoint, and Inverse of a Matrix {Scope as in Chapter 4 (4.1 to 4.6) of (B) Part-I}. Introduction to Statistics, Measures of Dispersion, Range, Mean Deviation, Variance, and Standard Deviation, Analysis of Frequency Distribution {Scope as in chapter 15 of (A)}.

Unit – III

Linear Inequalities – Introduction, Algebraic Solution of Linear Inequalities in one variable and their Graphical Representation, Graphical Solution of Linear Inequalities in two variables, Solution of System of Linear Inequalities in two variables, Permutation, and Combinations {Scope as in chapters 6, 7 of (A)}.

Essential Textbooks

1. Mathematics; (A) Textbook for class XI, NCERT, 2006
2. Mathematics; (B) Textbook for class XII, NCERT (Part I and Part II), 2006

SEMESTER-II
MDC-MAT-201: BASIC MATHEMATICS – II

Credits = 3 (L = 2 + T = 1)
1 Credit of Lecture = 1 hr. and 1 Credit of Tutorial = 3 hrs.
So, 3 Credits = 2 hrs. of lectures and 3 hrs. of Tutorials in a week = 5 hrs. / week
Total Contact hours in a semester of 15 weeks = 75 hrs. (30 hrs. Lectures + 45 hrs. Tutorials)
Exam Time: 3 hours **Max. Marks:** 75
(68 External + 7 Internal Assessment)

Note:

1. The syllabus has been split into three Units: Unit – I, Unit – II, and Unit – III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 7 parts of 2 marks each, covering the entire syllabus. Two questions, each of 18 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.

Objective:
This course is designed to introduce the fundamental concepts of limits, continuity, differentiation, integration of functions of one variable, and Probability. Its objective is to acquaint students with various applications of these topics relating to extreme value problems and problems related to Probability.

Learning outcomes:
The course will enable the students to learn the following:

1. Idea of limit, continuity, derivative, and application of derivatives (increasing and decreasing of function, maximum, minimum, Rolle's Theorem).
2. Integration as an anti-derivative and the process of integration.
3. Basic Idea of probability.

Unit – I

Limits, continuity, Derivative, and differentiability (Excluding Trigonometric Functions). Introduction, intuitive Idea of derivative, Limits, Continuity, Differentiability. Exponential and Logarithmic Differentiation, Derivative of a function in a parameter, Second-order derivative. {Scope as in Chapter 13 of (A) & Chapter 5 Part-I of (B)}

Unit – II

Application of derivatives: Increasing and decreasing functions, Maxima and Minima, Rolle's Theorem, and Lagrange's Mean Value Theorem (without proofs, only geometric interpretation, and problems). {Scope as in Chapter 6 of (B) Part-I}

Unit – III

Integral Calculus – (Excluding Trigonometric Functions). Integral as an anti-derivative, Integration by substitution, by partial fractions, and by parts. {Scope as in Chapters 7 & 8 of Part II of (B)}. Probability – Introduction, Random Experiments, Event, Axiomatic Approach to Probability {Scope as in Chapter 16 of (A)}.

Essential Textbooks

1. Mathematics, (A) Textbook for class XI, NCERT, 2006
2. Mathematics, (B) Textbook for class XII, NCERT (Part I and Part II), 2006

Microbial Food Technology

SEMESTER-I

SPECIAL FOODS AND MEAL MANAGEMENT

(CREDITS: 3)

MAX. MARKS:
THEORY:
INTERNAL ASSESSMENT:

INSTRUCTIONS FOR THE EXAMINER:
The Question Paper will have four sections. The examiner will set a total of nine questions, comprising two questions from each unit, and one compulsory question of short-answer type covering the whole syllabus. Students will attempt one question from each unit and the compulsory question. All questions may carry equal marks unless specified.

COURSE OBJECTIVES AND LEARNING OUTCOMES:

- To understand the scope of specialty foods.
- To classify specialty foods on the basis of sources.
- To familiarize with the concept of therapeutic foods.
- To gain insights into consumer requirements in the area of specialty foods.

UNIT I

No. of Hours: 07

Need and Scope of Specialty Foods: Need and scope of specialty foods, Functional foods, Convenience food, Healthcare and medical benefits, Nutritional status, Low-cost foods.
Specialty Foods Based on Sources: Cereals and millets, Legumes and pulses, Fruits and vegetables, Animal food sources, By-product based foods, Bioactive components, Novel nutraceuticals products, Fast foods, Organic and inorganic farming, Genetically modified foods.

UNIT II

No. of Hours: 08

Introduction to Human Nutrition: Introduction to human nutrition, Energy value of foods and from proximate principles, Daily calorie needs for basal metabolism.
Dietary Requirements of Nutrients: Requirements and role of carbohydrates, lipids, water, proteins, vitamins, and minerals in human health, Recommended Dietary Allowance (RDA), Dietary sources, Anti-nutritional factors in plant foods.

UNIT III

No. of Hours: 08

Therapeutic Foods: Modification of diets in disorders of the digestive tract, liver, cardiovascular system, kidney, metabolic disorders, allergy, endocrine disorders.
Specific Consumer-Oriented Foods: Foods for Defense persons, Space/astronauts, High-altitude mountain climbers, Disaster situations – crises, care, and maintenance.

UNIT IV

No. of Hours: 07

Nutrition Management Techniques: Infant nutrition and infant foods, Geriatric nutrition, Maternal nutrition, Sports Nutrition.

Advances in Nutrition: Effect of processing, preservation, and storage on the nutritional quality of foods, Food fortification, Food enrichment, Complementation and supplementation diets, Nutritional labeling.

SUGGESTED READINGS (latest editions):

1. Gibson, G.R. & William, C.M. (2000). ***Functional Foods - Concept to Product***.
2. Robert, E.C. (2006). ***Handbook of Nutraceuticals and Functional Foods***, 2nd Ed., Wildman.
3. Manson, P. (2001). ***Dietary Supplements***, 2nd Ed., Pharmaceutical Press.
4. Bamji, M.S., Rao, N.P. & Reddy, V. (2003). ***Textbook of Human Nutrition***. Oxford & IBH.
5. B. Srilakshmi. (2014). ***Dietetics***, 8th Ed., New Age Publication.

SEMESTER-II

Functional Foods for Sustainable Health

(CREDITS-3)

TOTAL HOURS-30
MAX. MARKS:
INTERNAL ASSESSMENT:

INSTRUCTIONS FOR THE EXAMINER: The Question Paper will have four sections. The examiner will set a total of nine questions, comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus. Students will attempt one question from each unit and the compulsory question. All questions may carry equal marks unless specified.

COURSE OBJECTIVES AND LEARNING OUTCOMES: The course is designed to inculcate knowledge about functional foods, their importance, and health benefits. Further, the students will acquire knowledge on the health-promoting nutritional factors and bioactive constituents, their potential health implications, and mechanisms of action.

UNIT I

No. of Hours: 08

Functional Foods and Nutraceuticals and Role in Health Management:
Introduction to Functional Foods: Definition, History, and Classification; Introduction to Nutraceuticals: Definition, Health benefits; Applications of Nutraceuticals in health and wellness; Introduction to Probiotics, Prebiotics, and Synbiotics.

UNIT II

No. of Hours: 07

Probiotics, Prebiotics, and Synbiotics:
Probiotics - Taxonomy and Important Features of Probiotic Microorganisms; Quality Assurance of Probiotics and Safety; Probiotics in various foods and health benefits; Types of prebiotics and their health benefits.

UNIT III

No. of Hours: 07

Functional foods and their types:
Conventional functional foods (e.g., Millets); Modified functional foods (Fortification, Enrichment, and Supplementation); Foods for special dietary use (Weight loss foods, Gluten-free foods, Lactose-free foods).

UNIT IV

No. of Hours: 08

Bioactive Components in Foods and Health effects:
Introduction to bioactive compounds: Description, health effects, and food examples; Polyphenols, Flavonoids, Carotenoids; Anti-nutritional Factors, Phytates; Vitamins, Minerals; Special reference to Resveratrol, Quercetin, Cinnamaldehyde, Capsaicin, Piperine, Gingerol, Eugenol, Diosgenin.

SUGGESTED READINGS (latest editions):

1. Wildman, R. E. (2016). *Handbook of Nutraceuticals and Functional Foods.* CRC Press
2. Gibson, G. R. and Williams, M. C. (2001). *Functional Foods: Concept to Product.* CRC Press
3. Vattem, D. A. and Maitin, V. (2016). *Functional Foods, Nutraceuticals, and Natural Products: Concepts and Applications.* DEStech Publications, Inc.
4. Gupta, R. C. (2016). *Nutraceuticals: Efficacy, Safety, and Toxicity.* Academic Press
5. Robert E. C. Wildman; *Handbook of Nutraceuticals and Functional Foods, Second Edition*; CRC Press
6. Giuseppe Mazza; *Functional Foods: Biochemical and Processing Aspects, Volume 1*; CRC Press

Microbiology

Semester-I
MDC-MIC -1: Introduction to Microbiology

Total marks:50
Theory: 45
Internal assessment:05
Time: 3 hours
Credits: 02

Note: The question paper will consist of Two Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be four questions from each Unit and two is to be attempted from each unit

Objective and Learning outcome:
Aim of the course is to introduce the students with the microbes .Microbes play a significant role in medical science and industries .To study the microbes from basic, with an understanding of cell structure, and application of microbes.

UNIT–I

History and scope of microbiology
General introduction and overview of microbial world. Major division of life. Members of microbial world and their characteristics. The Morphology of Bacteria ,viruses, viroids and Prions, fungi, Algae, protozoa.The development of techniques for studying microbial pathogens. The scope and relevance of microbiology.

Microbiota and Host relationship
The distribution and role of Normal microflora of human Body. The role of microorganisms in diseases. Microbial interactions like symbiosis and antibiosis.

UNIT-II

MICROSCOPY
Principles of microscopy; resolution, magnification; compound microscope**,** Bright field microscopy, Dark field microscopy. The influence of environmental factors on growth of microorganisms.

Media Preparation
Culture media: Different types of media preparation methods. Sterilization and pasteurization; The Physical and chemical agents to kill microbes.

SUGGESTED READING

1. Willey, J. M., Sherwood, L. M., & Woolverton, C. J. (2014). *Prescott's microbiology*.McGraw-Hill.
2. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall
3. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
4. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
5. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.

6. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction (Benjamin/ Cummings publishing company, Inc).
7. Stanier R Y , M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillian press London).
8. Willey JM,Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Postgate, J.R. (2000). Microbes and man (Cambridge University Press).
10. Alcamo, I. E. (2003). *Microbes and society: an introduction to microbiology*. Jones & Bartlett Learning.
11. Cappucino J and Sherman N.(2010).Microbiology:ALaboratoryManual9th edition.Pearson Publication

MDC-MIC -1: Introduction to Microbiology (Practical)

Total: 25
Practical: 22
INTERNAL ASSESSMENT:03
TIME: 3 hours
CREDITS:1

List of the Practical:

1. Basic Laboratory Rules and Instruments used in microbiology lab.
2. The components, use, care and measurements of microorganisms of the compound Microscope.
3. Sterilization of items and Preparation of culture media.
4. Microscopic examination of microorganism by Temporary Wet Mount.
5. Demonstration of Motility by hanging drop method.
6. Demonstration of Normal Microbial flora of the skin.
7. Dilution and pour plating techniques.

Semester-II

MDC-MIC-2: Tools and Techniques in Microbiology

Total marks: 50
Theory: 45
Internal assessment: 05
Time: 3 hours
Credits: 02

Objective: To enable the students learn important tools and techniques in microbiology so that they can understand application of these techniques.

Unit I

Principle, working and applications of Spectrophotometry (UV & visible)
Electrophoresis: Principle, types & applications.

Unit II

Microscopy: Properties; Light and Bright field Microscopy, Dark field Microscopy; Phase contrast, Fluorescent microscopy, Electron microscopy: Scanning and Transmission Electron Microscopy.

Unit III

Overview of chromatography; ion exchange chromatography, gel filtration chromatography

Unit IV

Radioisotope techniques: Radiotracers; GM counter, proportional and scintillation counters, autoradiography

MDC-MIC -2: Tools and Techniques in Microbiology (Practical)

Total marks:25
Practical:22
Internal assessment:03
Time:3 hours
Credits: 01
List of Experiments:

1. Quantitative analysis by UV/ Visible spectrophotometry.
2. Agarose gel electrophoresis
3. Paper chromatography
4. Gel filtration chromatography
5. Parts and working of microscope

SUGGESTEDREADINGS

1. K Wilson, J Walker (2000) Principles and techniques of biochemistry and Molecular Biology Seventh Edition, (Cambridge University Press).
2. SR Mikkelsen, E Cortin (2004). Bio analytical Chemistry(2004) (John Wiley&Sons)
3. David Sheehan (2009). Physical Biochemistry: Principles and Applications, Second Edition. (John Wiley & Sons)
4. P. Carmona, R. Navarro, A, Hernanz (1997) Spectroscopy of biological molecules: Modern trends (Illustrated, Publisher Springer)
5. Bernard Valeur (2001). Molecular fluorescence: Principles and application (Wiley-VCH).Plenum USA].

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Physics

The college has to opt any one course in each semester (Semester 1 and 2) out of the following courses:

Multidisciplinary Course (MDC)-1 (Exploring Physics in daily life)	
Outline Syllabus	**Exploring Physics in Daily Life**
Credit	**3**
Contact Hours	**45**
Marks	**75(Theory=60+ Internal Assessment =15)**
Instructions	1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration. 2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of short answer type questions covering the whole syllabus. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III. 3. Use of non-programmable calculators will be allowed in the examination centre.
Syllabus	**Unit 1: Physics and Technology** Physics of Communication and Information Technology • Basics of electromagnetism and its role in communication • Electric and magnetic fields • Electromagnetic waves and their properties • Transmission and reception of signals • Principles of signal processing and data transmission • Analog and digital signals • Modulation and demodulation techniques • Error detection and correction • Application of physics in electronic devices • Semiconductor physics and diode operation • Transistors and amplifiers • Integrated circuits and digital logic Physics in Transportation • Fundamentals of motion and forces in transportation • Newton's laws of motion • Velocity, acceleration, and force • Friction and drag forces • Analysis of automotive systems using physics principles • Engine operation and thermodynamics • Braking systems and energy dissipation • Suspension and shock absorbers • Understanding aerodynamics and its impact on vehicle design • Air resistance and drag coefficients • Lift and the Bernoulli principle • Shape optimization for efficiency

	Unit 2: Physics and Everyday Phenomena Energy and Sustainable Living • Introduction to energy forms and conservation • Energy transformations and the laws of thermodynamics • Energy efficiency and energy conservation techniques • Renewable and non-renewable energy sources • Exploring renewable energy sources • Solar energy and photovoltaic cells • Wind energy and turbine technology • Hydroelectric power and tidal energy Physics-based concepts for sustainable living and environmental impact • Carbon footprint and greenhouse gas emissions • Energy-efficient buildings and appliances • Sustainable transportation options Physics in Everyday Phenomena • Physics in household applications • Understanding simple machines and their everyday applications • Basics of electricity and electrical devices used in households Energy-efficient practices for everyday appliances

References

1. "Fundamentals of Physics" by David Halliday, Robert Resnick, and Jearl Walker
2. "Introduction to Electrodynamics" by David J. Griffiths
3. "Semiconductor Physics and Devices" by Donald A. Neamen
4. "Automotive Technology: Principles, Diagnosis, and Service" by James D. Halderman
5. "Aerodynamics for Engineers" by John J. Bertin and Russell M. Cummings
6. "Physics for Scientists and Engineers" by Paul A. Tipler and Gene Mosca
7. "Renewable Energy: Power for a Sustainable Future" by Godfrey Boyle
8. "Physics for Entertainment" by Yakov Perelman
9. "Energy: Its Use and the Environment" by Roger A. Hinrichs and Merlin H. Kleinbach.

Multidisciplinary Course (MDC)-2 Physics of Vacuum and Low Temperature	
Outline Syllabus	**Physics of Vacuum and Low Temperature**
Credit	**3**
Contact Hours	**45**
Marks	**75(Theory=60+ Internal Assessment =15)**
Instructions	1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration. 2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of short answer type questions covering the whole syllabus. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III. 3. Use of non-programmable calculators will be allowed in the examination centre.
Syllabus	**Unit I** Basics of Vacuum Techniques: Introduction, classification of vacuum ranges, throughput, Pump speed, speed of exhaust, conductance, ultimate pressure, viscous flow, molecular flow. Production of Low Pressures: Pump types, Gaede oil-sealed rotating vane pump, Diffusion pump, sputter ion pumps, Gettering, types of getters, Cryogenic pumps. **Unit II** Measurement of Low Pressures: Types of gauges, Mcleod gauge, Pirani gauge, Measurement of ultrahigh vacuum, penning gauge. Methodology of Vacuum systems: Vacuum system, Materials for vacuum system, cleaning and sealing of vacuum system, Leak detection and its location.

References

1. Vacuum Technology: A. Roth (North Holland, 1990).
2. Vacuum Technology: A. Guthrie (John Wiley, 1963).
3. A Treatise on Heat: M.N. Saha and B.N. Srivastava (Indian Press, 1965).
4. Low Temperature Physics: C. Dewitt, B. Dreyfus and P.G. de Gennes (Gordon & Breach, 1962).

Multidisciplinary Course (MDC) -3	
Outline Syllabus	**Computational Physics**
Credit	**3**
Contact Hours	**45**
Marks	**75 (Theory=60+ Internal Assessment =15)**
Instructions	1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration. 2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of short answer type questions covering the whole syllabus. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III. 3. Use of non-programmable calculators will be allowed in the examination centre.
Syllabus	**Unit I** **Basics of scientific computing**: Binary and decimal arithmetic, Floating point numbers, algorithms, Sequence, Selection and Repetition, single and double precision arithmetic, underflow & overflow- emphasize the importance of making equations in terms of dimensionless variables, Iterative methods. **Review of C & C++ Programming**: Introduction to Programming, constants, variables fundamentals and data types, operators and Expressions, I/O statements, scanf and printf, c in and c out, Manipulators for data formatting, Control statements (decisionmakingandlooping statements) (If statement. If else Statement. Nested if Structure. Else-if Statement. Ternary Operator. Goto Statement. Switch Statement. Unconditional and Conditional Looping. WhileLoop. Do-While Loop. FOR Loop. Break and Continue Statements. Nested Loops), Arrays (1D &2D) and strings, user defined functions, Structures and Unions, Idea of classes and objects. **Unit II** Programs: Sum & average of a list of numbers, largest of agiven list of numbers and its location in the list, sorting of numbers in ascending descending order, Binary search Random number generation Area of circle, area of square, volume of sphere, value of pi (π), Solution of linear and quadratic equation. Solution of first order differential equation, Radioactive decay, Classical equations of motion

Books:

1. Schaum's Outline of Programming with C++. J. Hubbard, 2000, McGraw□Hill Pub.
2. Numerical Recipes in C: The Art of Scientific Computing, W.H. Pressetal, 3 rdEdn. , 2007, Cambridge University Press.
3. An Introduction to Computational Physics, T.Pang, 2 ndEdn. , 2006,Cambridge Univ. Press.
4. Computational Physics, Darren Walker, 1st Edn., 2015, Scientific International Pvt. Ltd.

Multidisciplinary Course (MDC)-4	
Outline Syllabus	**ELEMENTS OF MODERN PHYSICS**
Credit	**3**
Contact Hours	**45**
Marks	**75(Theory=60+ Internal Assessment =15)**
Instructions	1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration. 2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of short answer type questions covering the whole syllabus. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III. 3. Use of non-programmable calculators will be allowed in the examination centre.
Syllabus	**Unit I** Planck's constant and light as a collection of photons; Photo-electric effect and Compton scattering. De Broglie wavelength and matter waves; Davisson-Germer experiment. Problems with Rutherford model- instability of atoms and observation of discrete atomic spectra; Bohr's quantization rule and atomic stability; calculation of energy levels for hydrogen like atoms and their spectra. Wave-particle duality, Heisenberg uncertainty principle- impossibility of a particle following a trajectory; Estimating minimum energy of a confined particle using uncertainty principle; Energy-time uncertainty principle. Schrodinger equation for non-relativistic particles; Momentum and Energy operators; stationary states; physical interpretation of wavefunction, probabilities and normalization; Probability and probability current densities in one dimension. One dimensional infinitely rigid box- energy eigenvalues and eigenfunctions, normalization; Quantum dot as an example; Quantum mechanical scattering and tunnelling in one dimension - across a step potential and across a rectangular potential barrier. **Unit II** Size and structure of atomic nucleus and its relation with atomic weight; Impossibility of an electron being in nucleus as a consequence of the uncertainty principle. Nature of nuclear force, NZ graph, semi-empirical mass formula and binding energy. Radioactivity: stability of nucleus; Law of radioactive decay; Mean life and half-life; α decay; β decay - energy released, spectrum and Pauli's prediction of neutrino; γ-ray emission. Fission and fusion - mass deficit, relativity and generation of energy; Fission - nature of fragments and emission of neutrons. Nuclear reactor: slow neutrons interacting with Uranium 235; Fusion and thermonuclear reactions.

Books:

1. Concepts of Modern Physics, Arthur Beiser, 2009, McGraw-Hill.
2. Modern Physics, J.R. Taylor, C.D. Zafiratos, M.A. Dubson,2009, PHI Learning
3. Six Ideas that Shaped Physics:Particle Behave like Waves, Thomas A. Moore, 2003, McGraw Hill
4. Quantum Physics, Berkeley Physics,Vol.4. E.H. Wichman, 2008, Tata McGraw-Hill Co.
5. Modern Physics, R.A. Serway, C.J. Moses, and C.A.Moyer, 2005, Cengage Learning
6. Modern Physics, G. Kaur and G.R. Pickrell, 2014, McGraw Hill.

Multidisciplinary Course (MDC) -5	
Outline Syllabus	**BASICS OF MATERIALS SCIENCE**
Credit	**3**
Contact Hours	**45**
Marks	**75(Theory=60+ Internal Assessment =15)**
Instructions	1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration. 2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of short answer type questions covering the whole syllabus. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III. 3. Use of non-programmable calculators will be allowed in the examination centre.
Syllabus	**Unit I** Review of Atomic Structure: Rutherford atom model, Bohr model of atom, Somerfield's Extension of Bohr's theory, Pauli's Exclusion principle, Wave mechanical model of the atom; Properties of crystalline and amorphous solids, crystal lattice, symmetry considerations, space lattice of cubic system, lattice planes and miller indices, Bragg's law, reciprocal lattice; Imperfection in crystals, point defects, dislocations, surface imperfections; Bonding in solids: Ionics bonding, covalent bonding, metallic bonding, van der Waal's bonding, Hydrogen bonding, Cohesive energy. Classical free electron theory of metals, Electrical conductivity, thermal conductivity, Wiedemann-Franz law, Quantum theory of free electrons, Fermi level, Fermi energy, Fermi sphere, Density of states. **Unit II** Bloch theorem, Kronig-Penny model for the band formation; Semiconductors: Direct and Indirect semiconductors, Drift velocity, mobility, Carrier concentration in n- and p-type semiconductors, Hall effect, Carrier transport in semiconductors, Techniques for formation of p-n junction, characteristics of a p-n junction. Properties of Solids: Dielectric properties: Polar and non-polar solids, concept of dielectric polarization, Dielectric constant and polarizability, polarization processes, Dielectric loss, Applications of dielectric materials, Ferroelectricity, Antiferroelectricity, pyroelectricity, piezoelectricity; Magnetic properties of solids: Definition of magnetism, origin of magnetic moment, Diamagnetism, Paramagnetism, Ferromagnetism, Antiferromagnetic and Ferrimagnetism

Books:

1. Introduction to Solid State Physics, Charles Kittel, 8 th Edition, 2004, Wiley India Pvt. Ltd.
2. Elements of Solid-State Physics, J.P. Srivastava, 4 th Edition, 2015, Prentice-Hall of India
3. Introduction to Solids, Leonid V. Azaroff, 2004, Tata Mc-Graw Hill
4. Solid State Physics, N.W. Ashcroft and N.D. Mermin, 1976, Cengage Learning
5. Essentials of Solid State Physics, S.P. Kuila, 2016, NCBA Ltd.
6. Elementary Solid State Physics, M. Ali Omar, 1999, Pearson India
7. Solid State Physics, M.A. Wahab, 2011, Narosa Publications

STATISTICS

SEMESTER-I
PAPER: B.Sc. STAT-MDC-101

Credit: 3
L-3

Notes:

1. This course will be offered by the Department of Statistics to the students of other Departments (other than the Department of Statistics).
2. Students are to be engaged in 3 hours of teaching per week.

Paper Title with code	Lecture (L)/ Credit	Marks Distribution
B.Sc.STAT-MDC-101 STATISTICS-I	Lecture:3 Credit:3	Theory Max.Marks:75 Theory: 60 Internal Assessment : 15 Time:3 Hours

STATISTICS-I

Objectives:

This course is designed to shape the attitudes of learners regarding the field of statistics. This course is aimed at students who need a basic background in statistics and its application. It is to provide an understanding for the student on statistical concepts to include measurements of location and dispersion, probability, probability distributions, regression, and correlation analysis. It provides knowledge about basic probability theory, distributions, and its significance in the real world. To analyze univariate and bivariate data, discrete and continuous random variables.

Learning Outcomes: On completion of this course, students will be able to:

(i) Understand the importance and scope of Statistics.
(ii) Get familiar with various techniques used in the summarization and analysis of data.
(iii) Understand the nature of statistical data.
(iv) Understand about the types of correlations for bivariate and multivariate data and their uses in real-life problems.
(v) Highlight the concept of linear regression and its applications in real life.
(vi) Understand the significance of statistics and probability in the real world.

Note: There will be a total of nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit. The candidate will be required to attempt five questions in all, including the compulsory first question and two questions from each Unit.

Unit-I

Introduction: Definition, importance, and scope of statistics. Types of statistical data: primary and secondary data, qualitative and quantitative data, time series data, discrete and continuous data. Scales of measurement - nominal, ordinal, interval, and ratio. Frequency distributions, cumulative frequency distributions. Diagrammatic representation of data: Bar diagrams, histogram, pie chart, measures of central tendency, measures of dispersion, moments, skewness, kurtosis,
Bivariate Data: Definition, scatter diagram, simple correlation coefficient and its properties, rank correlation, simple linear regression, properties, fitting of linear regression equations using the principle of least squares, partial and multiple correlation coefficients in three variables (only results, no derivation).

Unit-II

Basic Concepts of Probability: Random experiment, sample space, events, different definitions of probability, additive law of probability, multiplication probability, conditional probability.
Random Variables (for univariate case only): Discrete and continuous random variables, probability density function, distribution functions, mathematical expectation, moment generating function.

SUGGESTED READINGS:

1. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2013): *Fundamentals of Statistics, Vol I*, World Press, Kolkata.
2. Miller, Irwin and Miller, Marylees (2006): *John E. Freund's Mathematical Statistics with Applications*, (7th Edn.), Pearson Education, Asia.
3. Gupta, S.C. & Kapoor, V.K.: *Fundamentals of Mathematical Statistics*, Sultan Chand and Sons.
4. Mood, A.M., Graybill, F.A., and Boes, D.C. (2011): *Introduction to the Theory of Statistics*, 3rd Edn., (Indian Edition), Tata McGraw-Hill Pub. Co. Ltd.

SEMESTER-II
PAPER: B.Sc. STAT-MDC-102

Credit: 3
L-3

Notes:

1. This course will be offered by the Department of Statistics to the students of other Departments (other than the Department of Statistics).
2. Students are to be engaged to 3 hours of teaching per week.

Paper Title with code	Lecture (L) / Credit	Marks Distribution
B.Sc.STAT-MDC-102 STATISTICS-II	Lecture:3 Credit:3	Theory Max.Marks:75 Theory: 60 Internal Assessment : 15 Time:3 Hours

STATISTICS-II

Objectives:
The purpose of this course is to introduce students to the application of statistical methods and to make inferences about populations. Among the statistical methods that students will be exposed to are the following: t-tests, F-distribution, Chi-square distribution, and one-way and two-way ANOVA. It provides knowledge to perform hypothesis tests on means and proportions for one or two populations for small as well as large samples.

Learning Outcomes: On successful completion of the course, students will be able to:

- Understand concepts of sample, population, statistical hypotheses and their significance in the real world.
- Analyze and perform tests based on t, F, chi-square, and normal variate z for one and two samples.
- Perform one-way and two-way analysis of variance.
- Form tests of hypotheses as well as calculate confidence intervals for a population parameter for single-sample and two-sample cases.

Note: There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short-answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit. The candidate will be required to attempt five questions in all, including the compulsory first question and two questions from each Unit.

Unit-I

Concept of population and sample, parameter and statistic, need for sampling, types of sampling (definition only), sampling distribution of a statistic, standard error, statistical hypothesis, null and alternative hypotheses, simple and composite hypotheses, procedure in hypothesis testing, types of errors, power of test and critical region, levels of significance, one and two-tailed tests, degrees of freedom, Tests of hypothesis - large sample tests: test of hypothesis for single mean, difference of means, proportions, difference of proportions.

Unit-II

Sampling distributions and their applications: Student - t distributions, F-distribution, and Chi-square distribution. Simple tests based on t, F, chi-square, and normal variate z. Analysis of variance: for one-way classification, two-way classification (for fixed effect model only).

SUGGESTED READINGS:

1. Hogg, R.V., Tanis, E.A., and Rao, J.M. (2009): Probability and Statistical Inference, Seventh Edition, Pearson Education, New Delhi.
2. Gupta, S.C., & Kapoor, V.K.: Fundamentals of Applied Statistics, Sultan Chand and Sons.
3. Goon, A.M., Gupta, M.K., & Dasgupta, B. (2005): Fundamentals of Statistics, Vol. II, WordPress, Calcutta.
4. Meyer, P.L. (1970): Introductory Probability and Statistical Applications, Oxford & IBH Publishing, New Delhi.

Zoology

Semester-I

Course Code: _____
Course Title: Health and Hygiene

Total Hours= 45 (Theory=45) **Credits= 3 (L=3 + T=0 + P=0)**
Total Marks= 75 (Theory=75)
Theory Marks= 75 {60 (Theory examination) +15 (Internal assessment)}
Duration for final examination: 3 hours

Instructions For Paper Setter and candidate

Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 8 parts of 1.5 marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Theory

Course Outcomes:

- *The students will be able to develop clarity on key concepts related to health and hygiene.*
- *The students will able to develop strategies for the promotion of health and hygiene.*
- *The students will be able to appreciate different dimensions of health and role of proper nutrition.*
- *The students will be able to understand the importance of hygiene, personal and community health.*

UNIT 1

Nutrition

- Classification of foods -Carbohydrates, proteins, lipids, vitamins and minerals.
- Balanced diet and Malnutrition.
- Nutritional deficiencies and disorders.

UNIT 2

Health

- Definition of Health.
- Concept of Health & Diseases.
- Factors influencing Health.
- Immunity: Innate and Acquired Immunity.
- Basic concept of Vaccines and Allergy.

UNIT 3

Diseases

- Introduction to diseases, typesofdisease:Communicable and Non-Communicable diseases.
- Communicable Diseases:Causes, Symptoms and Prevention - Measles, polio, Influenza,Chikungunya,RabiesandAIDS.
- Non-Communicable diseases and their preventive measures – Hypertension, Coronary Heart diseases, Stroke andDiabetes.

UNIT 4

Hygiene

- Hygiene: Concept,Meaning, Principles,andImportance.
- StandardHygienePractices.
- TypesofHygiene:Personal,Food,andCommunity.
- Healthawareness, Food adulteration and inspections.

SUGGESTED READINGS

1. All India Institute of Hygiene and Public Health: http://aiihph.gov.in/healthin-india-2016/Page5of10
2. Dunn, C. L. & Pandaya D. D. (1925). Indian Hygiene and Public Health. Calcutta: ButterWorthCo.(India) Ltd.
3. Ministry of Ayush: http://ayush.gov.in/ Ministry of Health and Family Welfare:https://mohfw.gov.in/
4. Park,K.(2009).PreventiveandSocialMedicine.Jabalpur: M/sBanarsidasBhanot.
5. Park, W. H. (2019). Public Health and Hygiene: In Contributions by Eminent Authorities.Germany:Forgotten Books.
6. Sharma, M. Branscum, P. & Atri, A. (2014). Introduction to Community and Public Health.USA: JohnWiley&Sons.
7. SwachhBharatMission:http://swachhbharatmission.gov.in/sbmcms/index.htm
8. KeshavSwarnkar,Communityhealth –NRBrothersPublishersIndore
9. JatinV.Modi andRenjithS.Chawan.EssentialsofPublicHealthandSanitation–Part I-IV
10. Murray, C. J. L. and A.D. Lopez. (1996). The Global Burden Of Disease. World HealthOrganization.Park,J.E. andPark,K.TextbookofCommunity Healthfor Nurses.
11. SwaminathanS.Principlesof Nutrition andDietetics.
12. Personal Hygiene- objectives and advantages – Health and optimum health, factors affectinghealth
13. CommunityHygiene:IntroductiontoSocial,IndustrialandSchoolhygiene

Semester-II

Course Code: _____
Course Title: UNDERSTANDING HUMAN BODY

Total Hours= 45 (Theory=45) **Credits= 3 (L=3 + T=0 + P=0)**
Total Marks= 75 (Theory=75)
Theory Marks= 75 {60 (Theory examination) +15 (Internal assessment)}
Duration for final examination: 3 hours

Instructions For Paper Setter and candidate

Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 8 parts of 1.5 marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Theory

Course Outcomes:

- *The students will be able to understand Human body, Body functions and Life's Processes.*

UNIT 1

Digestive System
Alimentary canal, Digestive Glands, Digestion, Deficiency diseases.
Respiratory System
Respiratory tract, Respiratory organ, working of respiratory system, Diseases.

UNIT 2

Cardio Vascular System
Structure of Heart, Blood, Rh factor, Blood groups.
Excretory System
Structure of Kidney, Osmoregulation, Kidney Diseases.

UNIT 3

Skeletal System
Main parts of Skeleton and their bones, Types of Bones, Cartilage, Ligaments and Tendons.
Muscular System
Structure of Skeletal Muscle, Types of Muscle, Muscle contraction and Types of skeletal Muscle fibers.
Reproductive System
Male and Female reproductive system, their parts and functions.

UNIT 4

Nervous System
Parts of Nervous system, Structure of Neuron and Nerves.
Endocrine System
Major Glands and their Functions, Disorders.

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co., 1997.
2. Guyton, A.S.: Text Book of Medical Physiology, 14th edition, W.B. Saunders Company, 2020.
3. Pradeep's A Text Book of Biology (2024-25)ISBN: 978939564966 Pradeep Publication India
4. Modern's ABC Plus of Biology (Part I & II) Dr. BB Arora & Sabharwal Notebook, 1 January 2021
5. Human Physiology 2007 Stuart Fox (Author)Publisher : McGraw-Hill Education; 10th edition (16 February 2007)

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